

FORMALIZING ANGULAR STRUCTURE OF PARTICLE JETS

By

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ABSTRACT OF THE DISSERTATION

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Acknowledgments

A nice thank you...

Dedication

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Chapter 1

Introduction

1.1 The Standard Model

According to the Standard Model (SM), all matter results from interactions between fundamental particles: lepton and quarks (fermions), and mediators (bosons). Table 1.1 shows fermions of the standard model, their anti-particle equivalent would have all the signs reversed for the quantum numbers. For example, an anti-electron (e^+) and anti-electron-neutrino ($\bar{\nu}_e$) lepton number would flip to $(-1, 0, 0)$ and charges (Q s) would be $+1$ and 0 respectively.

Leptons				Quarks			
	l	Q	(L_e, L_μ, L_τ)	q	Q	$(F_d, F_u, F_s, F_c, F_b, F_t)$	
First generation	e^-	-1	$(1, 0, 0)$	d	$-1/3$	$(-1, 0, 0, 0, 0, 0)$	
	ν_e	0	$(1, 0, 0)$	u	$2/3$	$(0, 1, 0, 0, 0, 0)$	
Second generation	μ^-	-1	$(0, 1, 0)$	s	$-1/3$	$(0, 0, -1, 0, 0, 0)$	
	ν_μ	0	$(0, 1, 0)$	c	$2/3$	$(0, 0, 0, 1, 0, 0)$	
Third generation	τ^-	-1	$(0, 0, 1)$	b	$-1/3$	$(0, 0, 0, 0, -1, 0)$	
	ν_τ	0	$(0, 0, 1)$	t	$2/3$	$(0, 0, 0, 0, 0, 1)$	

Table 1.1: Three generations of leptons (left) and quarks (right).

The mediators/bosons exchanged between the other fundamental particles give rise to the fundamental interactions: electromagnetism from the photon (γ), weak force from $W^\pm$ and Z bosons, the gluon (g) for the strong force, and the Higgs boson (h) for imparting mass to the fundamental particles. Various interaction vertices possible in SM, that become the building blocks of all fundamental interactions are shown in 1.2.

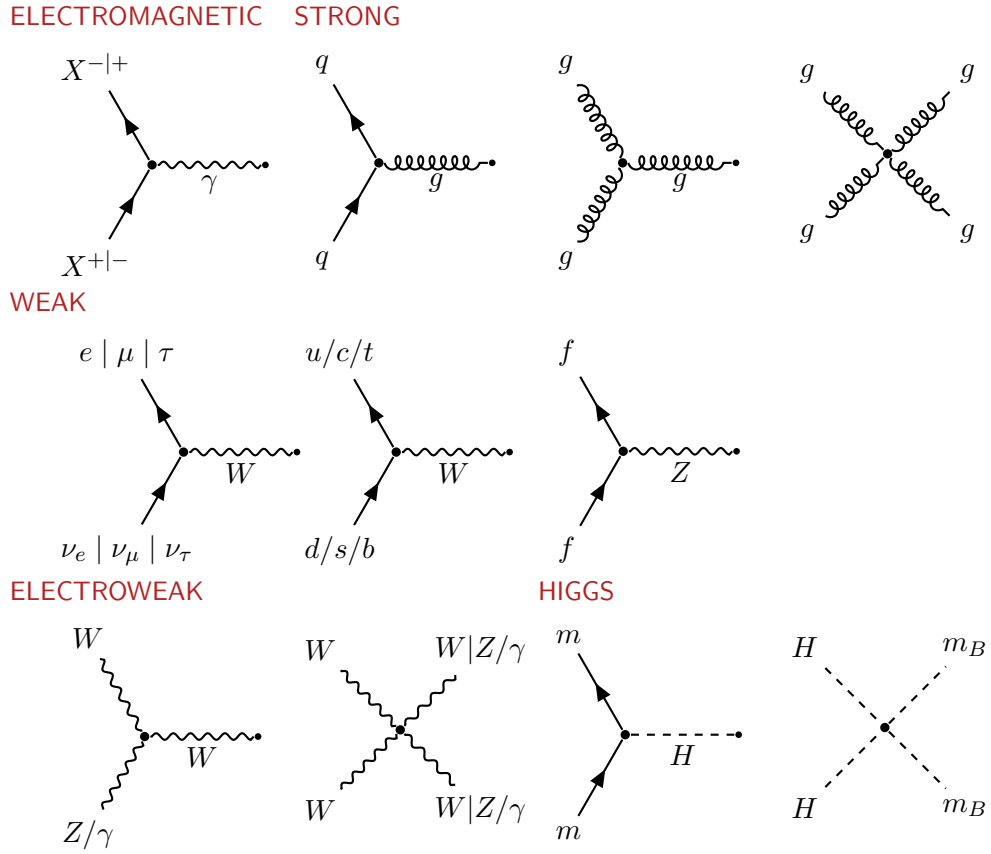


Table 1.2: Feynman diagrams of various interaction vertices possible in the Standard Model

1.2 Quantum Chromodynamics

Quantum Chromodynamics (QCD) is the component of the SM which describes the strong interactions of quarks and gluons. The QCD Lagrangian can be expected as:

$$\mathcal{L}_{\text{QCD}} = \underbrace{\bar{q}^\alpha (i\gamma^\mu \partial_\mu \delta_{\alpha\beta} - g\gamma^\mu t_{\alpha\beta}^a A_\mu^a - m\delta_{\alpha\beta}) q^\beta - \frac{1}{4} F_{\mu\nu}^a F_a^{\mu\nu}}_{\mathcal{L}_{\text{classical}}} - \underbrace{\frac{1}{2\lambda} (\partial^\mu A_\mu^a)^2}_{\mathcal{L}_{\text{gauge-fixing}}} + \underbrace{\bar{c}^\alpha \partial^\mu (\partial_\mu \delta_{\alpha\beta} + igT_{\alpha\beta}^a A_\mu^a) c^\beta}_{\mathcal{L}_{\text{ghost}}} \quad (1.1)$$

$\mathcal{L}_{\text{classical}}$ is the classical Lagrangian density shows interaction of a quark field q^α with 3 color degrees of freedom interacting with gluon fields A_μ^a with 8 color degrees of freedom. g and m are dimensionless and correspond to QCD coupling constant, and quark-mass respectively. $F_{\mu\nu}^a$ are the gluon tensors defined as,

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a - gf_{abc} A_\mu^b A_\nu^c. \quad (1.2)$$

To properly define gluon propagators, we need to make a choice of gauge. The $\mathcal{L}_{\text{gauge-fixing}}$ parametrizes the class of covariant gauges with the gauge parameter λ . In non-Abelian theories, this gauge fixing term leads to creation of unphysical degrees of freedom, which are canceled out by $\mathcal{L}_{\text{ghost}}$ which contains complex, scalar fermion fields c^α .

The matrices t^a and T^a are the fundamental and adjoint representations of SU(3) in form of traceless 3×3 and 8×8 hermitian matrices satisfying:

$$\begin{aligned} [t^a, t^b] &= if_{abc} t^c, \quad \text{tr}(t^a t^b) = \frac{1}{2} \delta^{ab} \\ [T^a, T^b] &= if_{abc} T^c, \quad (T^a)_{bc} = -if^{abc}. \end{aligned} \quad (1.3)$$

Comparing with quantum electrodynamics (QED), where the photon field tensor lacks the last term of Eqn. 1.2 as it is an Abelian, $U(1)$ gauge theory, we can explain the presence

of three/four-gluon vertices and absence of such vertices for the photon in Table 1.2. The lagrangian can be used to formulate *Feynman rules* for calculating amplitude contributions from various propagators and vertices, which are shown in Tab. 1.3.

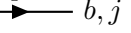
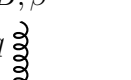
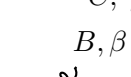
	$\delta^{AB} \left[-g^{\alpha\beta} + (1-\lambda) \frac{p^\alpha p^\beta}{p^2 + i\epsilon} \right] \frac{i}{p^2 + i\epsilon}$
	$\delta^{AB} \frac{i}{p^2 + i\epsilon}$
	$\delta^{ab} \frac{i}{(\not{p} - m + i\epsilon)_{ji}}$
	$-g f^{ABC} [(p-q)^\gamma g^{\alpha\beta} + (q-r)^\alpha g^{\beta\gamma} + (r-p)^\beta g^{\gamma\alpha}]$ (all momenta incoming, $p+q+r=0$)
	$-ig^2 f^{XAC} f^{XBD} [g^{\alpha\beta} g^{\gamma\delta} - g^{\alpha\delta} g^{\beta\gamma}]$ $-ig^2 f^{XAD} f^{XBC} [g^{\alpha\beta} g^{\gamma\delta} - g^{\alpha\gamma} g^{\beta\delta}]$ $-ig^2 f^{XAB} f^{XCD} [g^{\alpha\gamma} g^{\beta\delta} - g^{\alpha\delta} g^{\beta\gamma}]$
	$g f^{ABC} q^\alpha$
	$-ig (t^A)_{cb} (\gamma^\alpha)_{ji}$

Table 1.3: Feynman rules for QCD in a covariant gauge with gauge parameter λ : the gluon, ghost and quark propagators, followed by the three- and four-gluon, ghost-gluon and quark-gluon vertices.

1.2.1 The Running Coupling

At an energy scale Q , consider a dimensionless observable X . Lets assume that Q is sufficiently high that the quark masses become negligible in comparison. To calculate X as

a perturbation series in the QCD coupling $\alpha_s = \frac{g^2}{4\pi}$ (analogous to the QED fine structure constant), we need to remove ultraviolet divergences through renormalization. Renormalization introduces another arbitrary energy scale μ where the divergences are subtracted. In general, X would depend on the ratio Q^2/μ^2 and post-renormalization coupling $\alpha_s(\mu^2)$. However, any physical observable should be independent of μ :

$$\begin{aligned} \mu^2 \frac{dX(Q^2/\mu^2, \alpha_s(\mu^2))}{d\mu^2} &= \left[\mu^2 \frac{\partial}{\partial \mu^2} + \mu^2 \frac{\partial \alpha_s(\mu^2)}{\partial \mu^2} \frac{\partial}{\partial \alpha_s} \right] X(Q^2/\mu^2, \alpha_s(\mu^2)) = 0 \\ \implies \left[-\frac{\partial}{\partial t} + \beta(\alpha_s) \frac{\partial}{\partial \alpha_s} \right] X(e^t, \alpha_s) &= 0. \end{aligned} \quad (1.4)$$

Where, $t = \ln \frac{Q^2}{\mu^2}$, $\beta(\alpha_s) = \mu^2 \frac{\partial \alpha_s}{\partial \mu^2}$, $\alpha_s \equiv \alpha_s(\mu^2(t))$

$$\begin{aligned} \beta(\alpha_s) = -\frac{\partial \alpha_s}{\partial t} \implies t &= \int_{\alpha_s}^{\alpha_s(Q^2)} \frac{d\alpha_s}{\beta(\alpha_s)} \\ \implies \frac{\partial \alpha_s(Q^2)}{\partial t} &= Q^2 \frac{\partial \alpha_s}{\partial Q^2} = \beta(\alpha_s(Q^2)), \quad \frac{\partial \alpha_s(Q^2)}{\partial \alpha_s} = \frac{\beta(\alpha_s(Q^2))}{\beta(\alpha_s)} \end{aligned} \quad (1.5)$$

Substituting in Eq. 1.4, we see that $X(1, \alpha_s(Q^2))$ is a valid solution. Thus the scale dependence in X comes from the *running coupling* $\alpha_s(Q^2)$. Perturbatively expanding $\beta(\alpha_s)$, we can write the *renormalization group equation* as,

$$Q^2 \frac{\partial \alpha_s}{\partial Q^2} = -\alpha_s^2(Q^2) (\beta_0 + \beta_1 \alpha_s(Q^2) + \mathcal{O}(\alpha_s^2(Q^2))) \quad (1.6)$$

Where,

$$\beta_0 = \frac{1}{(4\pi)^2} \left(11 - \frac{2}{3} N_f \right), \quad \beta_1 = \frac{1}{(4\pi)^4} \left(102 - \frac{38}{3} N_f \right) \quad (1.7)$$

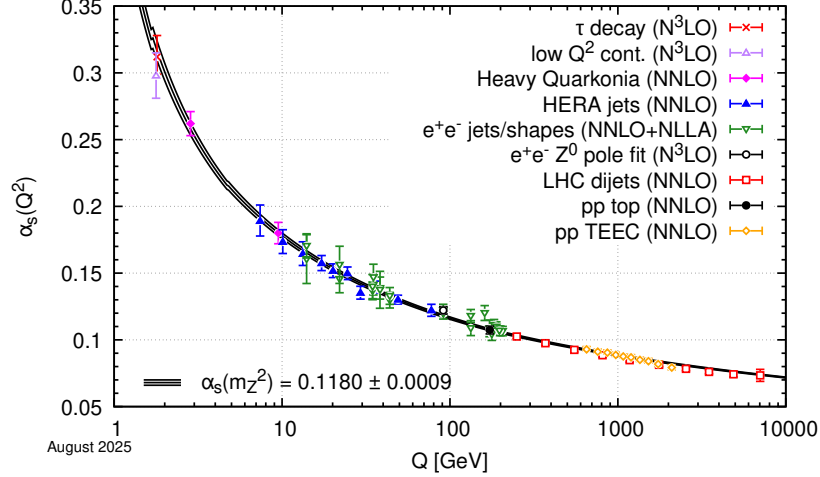


Figure 1.1: Summary of determinations of α_s as a function of the energy scale Q compared to the running of the coupling computed at five loops taking as an input the current Particle Data Group (PDG) world average, $\alpha_s(m_Z^2) = 0.1180 \pm 0.0009$ [1]

Integrating Eq. 1.6

$$\ln\left(\frac{Q^2}{Q_0^2}\right) = \int_{\alpha_s(Q_0^2)}^{\alpha_s(Q^2)} \frac{d\alpha_s}{\beta(\alpha_s)}$$

$$\Rightarrow \alpha_s(Q^2) = \frac{\alpha_s(Q_0^2)}{1 + \beta_0(\alpha_s(Q_0^2)) \ln(Q^2/Q_0^2) + \mathcal{O}(\alpha_s^2)} \quad (1.8)$$

Usually, the reference scale Q_0^2 is set to a convenient energy scale in the perturbative regime. α_s at all other scales can be inferred accordingly. Usually, Q_0 is set to mass of the Z -boson, m_Z as shown in Fig. 1.1. Another approach is to choose a scale parameter, Λ such that $\alpha_s(\Lambda) \rightarrow \infty$.

$$\ln\left(\frac{Q^2}{\Lambda^2}\right) = - \int_{\alpha_s(Q^2)}^{\infty} \frac{d\alpha_s}{\beta(\alpha_s)}$$

$$\Rightarrow \alpha_s(Q^2) = \frac{1}{\beta_0 \ln(Q^2/\Lambda^2)} \left[1 - \frac{\beta_1}{\beta_0} \frac{\ln(\ln(Q^2/\Lambda^2))}{\ln(Q^2/\Lambda^2)} + \mathcal{O}\left(\frac{1}{\ln(Q^2/\Lambda^2)^2}\right) \right] \quad (1.9)$$

Λ is called the *confinement scale*, around 200 MeV. Eqs. 1.8 and 1.9 show a monotonically decreasing relationship between α_s and Q^2 . At high momentum transfers ($Q \gg \Lambda$), the

coupling strength vanishes ($\alpha_s \rightarrow 0$). In this regime, quarks and gluons behave as quasi-free particles, making perturbative QCD (pQCD) calculations highly reliable (*Asymptotic Freedom*). At low energy scales ($Q \ll \Lambda$), the coupling diverges. In this non-perturbative QCD (npQCD) limit, isolated color charges cannot exist and partons are permanently bound into color-neutral hadrons (*Color Confinement*).

1.2.2 Quark Gluon Plasma (QGP)

Asymptotic freedom of QCD means that matter at extremely high temperatures and densities (high Q^2) undergo a *deconfinement* phase transition, with partonic degrees of freedom replacing hadronic degrees of freedom. This state of deconfined quarks and gluons is called QCD/quark-gluon plasma due to analogous phenomena observed in atomic physics. Thermodynamically, when temperature (T) or baryon number density (n_B) approaches the confinement scale Λ introduced in Sec. 1.2.1 ($T \sim \Lambda \sim \mathcal{O}(10^{12})K$, and $n_B \sim \Lambda^3 \sim 1 \text{ fm}^{-3}$) the boundaries between hadrons become ill defined resulting in hadrons “melt” into QGP. For reference, the core of our sun “only” has $T \sim \mathcal{O}(10^7)K$ and $n_B \sim \mathcal{O}(10^{-14})\text{fm}^{-3}$.

These orders of T and n_B are the theorized conditions prevalent in the universe a few μs after Big-Bang. The only natural phenomena in the present-day, cold universe that can achieve these conditions are the cores of neutron stars. Artificially, the required T and n_B can be achieved at facilities like the Large Hadron Collider (LHC) and Relativistic Heavy Ion Collider (RHIC) through collisions of heavy-ions (like Au, Pb, etc.) at near-light speeds, the aptly named *little bangs*. More on this will be discussed in Sec. 3.2.

1.3 Hard Processes

To consider any nucleus as a singular coherent object, the fluctuations inside it would involve momentum transfers below the confinement scale (Λ) introduced in 1.2.1, and would occur over timescales $\sim 1/\Lambda$. We define *hard-scattering processes* or just *hard processes* as interactions of a hard perturbative probe with high momentum transfer $Q \gg \Lambda$. Since

this interaction occurs over a time-scale $\sim 1/Q \ll 1/\Lambda$, the perturbative probe effectively interacts with a near-static “snapshot” of the nucleus with a characteristic resolution $\sim 1/Q$.

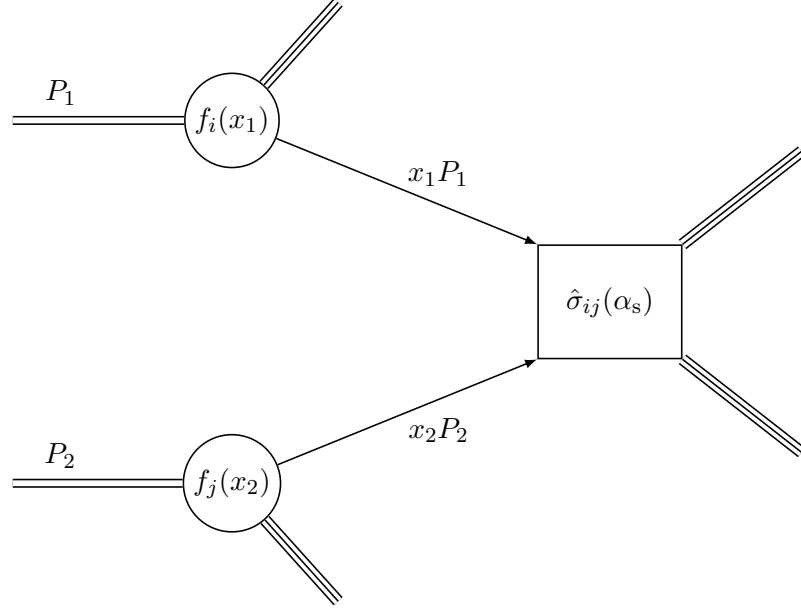


Figure 1.2: Collinear factorization of a hadronic hard scattering.

Thus, we can *factorize* the cross section of a collision process of nuclei with momentum P_1 and P_2 depicted by Fig. 1.2 as,

$$\sigma(P_1, P_2) = \sum_{i,j} \int dx_1 dx_2 f_i(x_1, \mu^2) f_j(x_2, \mu^2) \hat{\sigma}_{ij}(p_1, p_2, \alpha_s, Q^2/\mu^2) \quad (1.10)$$

Where, the interactions occur between partons i from nucleus 1 and j from nucleus 2 carrying momentum $p_1 = x_1 P_1$ and $p_2 = x_2 P_2$ respectively. μ is an arbitrary *factorization scale*, often just set to Q . The $f_k(x, \mu^2)$ are the *parton densities* giving likelihood of a parton k to carry x fraction of it's parent nucleus's momentum at a particular factorization scale μ . $\hat{\sigma}_{ij}$ is the *parton cross-section* for scattering of partons i and j . We can approximate it to the $(n+k)$ order in α_s as,

$$\hat{\sigma}_{ij}(p_1, p_2, \alpha_s, Q^2/\mu^2) = \alpha_s^k \sum_{m=0}^n \hat{\sigma}_{ij}^{(m)}(p_1, p_2, Q^2/\mu^2) \alpha_s^m \quad (1.11)$$

Different hard processes will have different leading powers k . W production will have $k = 0$ whereas perturbative dijet production with two outgoing quarks will have $k = 2$.

1.4 From partons to jets

Outgoing partons evolve from higher to lower momentum transfer scales (say, t). Initially their evolution are modeled using the pQCD mechanism called *parton shower*. Eventually, say at an infra-red scale cut-off t_0 , pQCD calculations stop being valid and npQCD processes like hadronization dominate. QCD Monte-Carlo programs combining parton shower at $t < t_0$ and hadronization process at $t > t_0$ are called *QCD event generators*

1.4.1 Parton branching

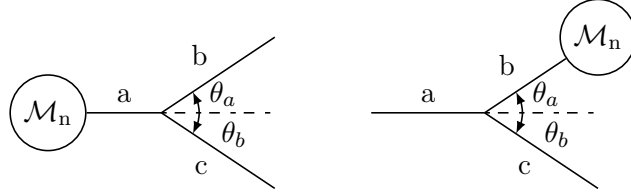


Figure 1.3: Kinematics of timelike (left) vs spacelike (right) parton branching

Consider timelike splitting of a parton, $a \rightarrow b + c$ shown in Fig. 1.4.1. Let's assume the partons are massless, and define the evolution scale (t) as,

$$t \equiv p_a^2 \gg p_b^2, p_c^2$$

$$\implies t \approx 2E_b E_c (1 - \cos(\theta)) \approx z(1-z)E_a^2 \theta^2 \quad (1.12)$$

Where, $z = E_b/E_a = 1 - E_c/E_a$ and $\theta = \theta_a + \theta_b$ is the opening angle. The cross-section of a n -parton final state branching processes can be calculated by considering n -parton matrix element.

$$d\sigma_n = \mathcal{F} |\mathcal{M}_n|^2 d\Phi_n \quad (1.13)$$

Where $\mathcal{F}$ is the initial-state flux factor and Φ_n is phase space of the n -parton final state.

In the branching process the n -parton pre-split phase space volume element:

$$d\Phi_n = \dots \frac{d^3\mathbf{p}_a}{2(2\pi)^3 E_a} \quad (1.14)$$

will be replaced by the $n + 1$ -parton phase space and the $n + 1$ -parton matrix element can be calculated from the n -parton matrix element.

$$\begin{aligned} d\Phi_{n+1} &= \dots \frac{d^3\mathbf{p}_b}{2(2\pi)^3 E_b} \frac{d^3\mathbf{p}_c}{2(2\pi)^3 E_c} = d\Phi_n \frac{1}{4(2\pi)^3} dt dz d\phi \\ |\mathcal{M}_{n+1}|^2 &\sim \frac{8\pi\alpha_s}{t} CF |\mathcal{M}_n|^2 \end{aligned} \quad (1.15)$$

Substituting in Eq. 1.13 and integrating away the $d\phi$, we get,

$$\begin{aligned} d\sigma_{n+1} &= d\sigma_n \frac{dt}{t} dz \frac{\alpha_s}{2\pi} \int \frac{d\phi}{2\pi} CF \\ &= d\sigma_n \frac{dt}{t} dz \frac{\alpha_s}{2\pi} \hat{P}_{ba} \end{aligned} \quad (1.16)$$

Same equation would hold for the space-like case. To obtain the evolution equations, we can frame the evolution in terms of the change in parton distribution function $f(x, t)$ when the evolution scale t goes from t to $t + \delta t$.

$$\begin{aligned} \delta f_i(x, t) &= \delta f_{\text{in}} - \delta f_{\text{out}} \\ &= \frac{\delta t}{t} \sum_j \int_0^1 dz \frac{\alpha_s}{2\pi} \left[\frac{\hat{P}_{ij}(z)}{z} f_j(x/z, t) - \hat{P}_{ji} f_i(x, t) \right] \end{aligned} \quad (1.17)$$

Introducing the *Sudakov form factor*, which gives the probability of a parton i evolving from scale t to a softer scale t_0 without a resolvable emission,

$$\Delta_i(t) \equiv \exp \left(- \sum_j \int_{t_0}^t \frac{dt'}{t'} \int dz \frac{\alpha_s}{2\pi} \hat{P}_{ji}(z) \right) \quad (1.18)$$

We get the familiar DGLAP-like evolution equations,

$$t \frac{\partial}{\partial t} \left(\frac{f_i}{\Delta_i} \right) = \frac{1}{\Delta_i} \sum_j \int \frac{dz}{z} \frac{\alpha_s}{2\pi} \hat{P}_{ij}(z) f_j(x/z, t) \quad (1.19)$$

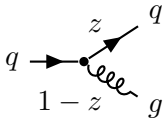
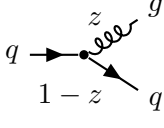
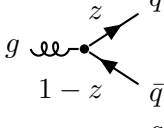
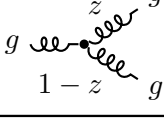
Kernel (ij)	Diagram	$\hat{P}_{ij}(z)$
qq		$C_F \left[\frac{1+z^2}{1-z} \right]$
gq		$C_F \frac{1+(1-z)^2}{z}$
qg		$T_R [z^2 + (1-z)^2]$
gg		$C_A \left[\frac{z}{1-z} + \frac{1-z}{z} + z(1-z) \right]$

Table 1.4: The four leading-order collinear splittings entering the DGLAP evolution of Eq. ?? . The daughter parton i carrying momentum fraction z is drawn on the upper leg, the recoiling parton $(1-z)$ on the lower. The plus prescription (Eq. ??) and $\delta(1-z)$ terms regularize the soft limit and enforce the momentum and flavor sum rules [3].

To deal with singularities corresponding to soft gluons at $z \rightarrow 1$, we introduce an explicit infrared cut-off $z < 1 - \epsilon(t)$. Requiring the transverse momentum in timelike branching

$a \rightarrow bc$ to be positive,

$$\begin{aligned}
p_T^2 &= z(1-z)p_a^2 - (1-z)p_b^2 - zp_c^2 > 0 \\
\implies z(1-z) &> t_0/t \\
\implies z, 1-z &> \epsilon(t) = \frac{1 - \sqrt{1 - 4t/t_0}}{2} \simeq t_0/t \\
\implies t_0/t &< z < 1 - t_0/t
\end{aligned} \tag{1.20}$$

$$\Delta_i(t) \equiv \exp \left(- \sum_j \int_{2t_0}^t \frac{dt'}{t'} \int_{t_0/t'}^{1-t_0/t'} dz \frac{\alpha_s}{2\pi} \hat{P}_{ji}(z) \right) \tag{1.21}$$

1.4.2 Angular Ordering

Sec. 1.4.1 took collinear enhancements into account to all orders in α_s . However it does not take the enhancement in soft gluon emission at wide emission angle. This is demonstrated in Eq. 1.22 by considering an on-mass-shell external line of momentum p , mass m emitting a gluon with momentum q at an angle θ

$$\frac{1}{(p \pm q)^2 - m^2} = \frac{\pm 1}{2p \cdot q} = \frac{\pm 1}{2\omega E(1 - v \cos \theta)}, \tag{1.22}$$

where ω is the gluon energy, E and v the energy and velocity of the emitter and θ the emission angle. For light partons $v \rightarrow 1$ and the collinear enhancement appears as $\theta \rightarrow 0$; the enhancement as $\omega \rightarrow 0$ holds for any velocity and angle. Internal lines are off-mass-shell and carry no such enhancement, since $(p + q)^2 - m^2 \rightarrow p^2 - m^2 \neq 0$. The cross section carries a sum over all pairs of external lines $\{i, j\}$,

$$d\sigma_{n+1} = d\sigma_n \frac{d\omega}{\omega} \frac{d\Omega}{2\pi} \frac{\alpha_s}{2\pi} \sum_{i,j} C_{ij} W_{ij} \tag{1.23}$$

The weighted sum $\sum_{i,j} C_{ij} W_{ij}$ is called the *antenna pattern* of the process. The *color factor* C_{ij} and *radiation function* W_{ij} are defined as,

$$C_{ij} = -\mathbf{Q}_i \cdot \mathbf{Q}_j$$

$$W_{ij} = \frac{\omega^2 p_i \cdot p_j}{p_i \cdot q \, p_j \cdot q} = \frac{1 - v_i v_j \cos \theta_{ij}}{(1 - v_i \cos \theta_{iq})(1 - v_j \cos \theta_{jq})} \quad (1.24)$$

The vectors $\mathbf{Q}_i$ represents color charge of parton i .

$$\mathbf{Q}_i^2 = \begin{cases} C_F = 4/3 & i \in \text{quarks} \\ C_A = 3 & i \in \text{gluons} \\ 0 & i \in \text{color singlet} \end{cases} \quad (1.25)$$

W_{ij} comes from interference between lines i and j . W_{ij} can be separated into two parts $W_{ij}^{[i/j]}$ each with an *incoherent* and a *coherent* contribution from line i/j . Assuming massless emitters ($v_i = v_j = 1$) for simplicity,

$$W_{ij} = W_{ij}^{[i]} + W_{ij}^{[j]}$$

$$W_{ij}^{[i]} = \frac{1}{2} \left(\overbrace{\frac{1}{1 - \cos \theta_{iq}}}^{\text{incoherent}} + \overbrace{\frac{\cos \theta_{iq} - \cos \theta_{ij}}{(1 - \cos \theta_{jq})(1 - \cos \theta_{iq})}}^{\text{coherent}} \right)$$

$$= \frac{1}{2(1 - \cos \theta_{iq})} \left(1 + \frac{\cos \theta_{iq} - \cos \theta_{ij}}{1 - \cos \theta_{ij} \cos \theta_{iq} - \sin \theta_{ij} \sin \theta_{iq} \cos \phi_{iq}} \right) \quad (1.26)$$

Using the direction of line i as the z-axis, $d\Omega$ from Eq. 1.23 becomes $d \cos \theta_{iq} d\phi_{iq}$, We can carry out the ϕ_{iq} integration,

$$\int_0^{2\pi} \frac{d\phi_{iq}}{2\pi} W_{ij}^{[i]} = \frac{1}{1 - \cos \theta_{iq}} \left(\frac{1 + \frac{\cos \theta_{iq} - \cos \theta_{ij}}{|\cos \theta_{iq} - \cos \theta_{ij}|}}{2} \right)$$

$$= \begin{cases} \frac{1}{1 - \cos \theta_{iq}} & \cos \theta_{iq} > \cos \theta_{ij} \implies \theta_{iq} < \theta_{ij} \\ 0 & \text{otherwise,} \end{cases} \quad (1.27)$$

Eq. 1.27 shows that azimuth-averaged soft gluon emission from a line i is confined to a

cone, centered on i 's direction, extending up to the direction of line j , i 's sibling from the preceding split. The total soft gluon emission cross-section can be then easily obtained for two external lines forming a color singlet ($e^+e^- \rightarrow q\bar{q}$) have $\mathbf{Q}_i + \mathbf{Q}_j = 0$ and hence $C_{q\bar{q}} = \mathbf{Q}_q^2 = \mathbf{Q}_{\bar{q}}^2 = C_F$, and angular ordering suppresses radiation outside the cones extending from q to $\bar{q}$ and vice versa.

$$d\sigma_{q\bar{q}g} = d\sigma_{q\bar{q}} \frac{\Theta(\cos(\theta_{qg}) - \cos(\theta_{q\bar{q}}))}{1 - \cos\theta_{qg}} d(\cos\theta_{qg}) \frac{\alpha_s}{2\pi} \frac{d\omega}{\omega} C_F \quad (1.28)$$

We will come back to Eq. 1.28

1.4.3 Interference

Lets take the case of three partons $\{i, j, k\}$ forming a color singlet ($\mathbf{Q}_i + \mathbf{Q}_j + \mathbf{Q}_k = 0$) e.g., $e^+e^- \rightarrow q\bar{q}g$. First, partons l and k come from a color singlet state ($\mathbf{Q}_l + \mathbf{Q}_k = 0$). Then l splits into i and j ($\mathbf{Q}_l = \mathbf{Q}_i + \mathbf{Q}_j = -\mathbf{Q}_k$). The antennae function for this state is:

$$\begin{aligned} \sum_{a,b} C_{ab} W_{ab} &= -\mathbf{Q}_i \cdot \mathbf{Q}_j W_{ij} - \mathbf{Q}_j \cdot \mathbf{Q}_k W_{jk} - \mathbf{Q}_i \cdot \mathbf{Q}_k W_{ik} \\ &= \frac{1}{2} [\mathbf{Q}_i^2 (W_{ij} + W_{ik} - W_{jk}) + \mathbf{Q}_j^2 (W_{ij} + W_{jk} - W_{ik}) + \mathbf{Q}_k^2 (W_{ik} + W_{jk} - W_{ij})] \\ &= \mathbf{Q}_i^2 \left(W_{ij}^{[i]} + \tilde{W}_{jk}^{[i]} - \tilde{W}_{ik}^{[j]} \right) + \mathbf{Q}_j^2 \left(W_{ij}^{[j]} + \tilde{W}_{ik}^{[j]} - \tilde{W}_{jk}^{[i]} \right) \\ &\quad + \mathbf{Q}_k^2 \left(\tilde{W}_{jk}^{[i]} + \tilde{W}_{ik}^{[j]} + \frac{W_{ik}^{[k]} + W_{jk}^{[k]}}{2} \right) \end{aligned} \quad (1.29)$$

Where,

$$\tilde{W}_{jk}^{[i]} = \frac{W_{ik}^{[i]} - W_{ij}^{[i]}}{2}, \quad \tilde{W}_{ik}^{[j]} = \frac{W_{jk}^{[j]} - W_{ij}^{[j]}}{2} \quad (1.30)$$

are shifted radiation functions for likes i and j , with the collinear singularity canceled out.

This allows us to make certain useful approximations. Taking $\theta_{ij} \rightarrow 0$, we can approximate

the directions of i and j by l .

$$W_{ik}^{[k]} \simeq W_{jk}^{[k]} \simeq W_{lk}^{[k]}, \text{ and } \tilde{W}_{jk}^{[i]} \simeq \tilde{W}_{ik}^{[j]} \simeq W_{lk}^{[ij]}/2 \quad (1.31)$$

Where,

$$\tilde{W}_{lk}^{[ij]} = \begin{cases} W_{lk}^{[l]} & \theta_{lq} > \theta_{ij} \\ 0 & \theta_{lq} < \theta_{ij}, \end{cases} \quad (1.32)$$

$$W \simeq \mathbf{Q}_i^2 W_{ij}^{[i]} + \mathbf{Q}_j^2 W_{ij}^{[j]} + \mathbf{Q}_k^2 W_{lk}^{[k]} + \mathbf{Q}_l^2 \tilde{W}_{lk}^{[ij]}, \quad (1.33)$$

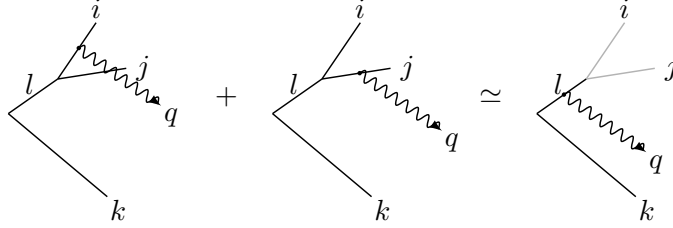


Figure 1.4: Wide-angle emission of soft gluon q , off i and j , acts as if the emission came off the parent parton l , imagined to be on shell.

Each parton radiates in proportion to its color charge squared. When i and j close in angle their incoherent contributions are limited to cones of half-angle θ_{ij} , while at larger angles, out to the direction of k , they radiate coherently in proportion to their combined charge $\mathbf{Q}_l^2$, as if from an internal line of momentum $p_l = p_i + p_j$ as shown in Fig. 1.4.

1.4.4 Coherent Parton Branching

Extended to higher orders this gives a *coherent parton branching* formalism. In place of the virtual mass-squared t the evolution variable is the opening angle, or more precisely

$$\zeta = \frac{p_b \cdot p_c}{E_b E_c} \simeq 1 - \cos \theta \quad (1.34)$$

for $a \rightarrow bc$, with angular ordering $\zeta' < \zeta$ imposed on successive branchings and the propagator factor dt/t of Sec. 1.4.1 replaced by $d\zeta/\zeta$. Using the splitting function $\hat{P}_{ba}(z)dz$ rather

than the soft approximation $\mathbf{Q}_a^2 d\omega/\omega$ treats both soft and collinear enhancements correctly, so that

$$d\sigma_{n+1} = d\sigma_n \frac{d\zeta}{\zeta} dz \frac{\alpha_s}{2\pi} \hat{P}_{ba}(z). \quad (1.35)$$

The virtual mass-squared cut-off is replaced by an angular one, $\zeta_0 = t_0/E^2$ for a parton of energy E , and the convenient evolution variable is

$$\tilde{t} = E^2 \zeta \geq t_0, \quad (1.36)$$

in terms of which the ordering condition $\zeta_b, \zeta_c < \zeta_a$ and the resulting cut-off on z read

$$\tilde{t}_b < z^2 \tilde{t}, \quad \tilde{t}_c < (1-z)^2 \tilde{t}, \quad \sqrt{t_0/\tilde{t}} < z < 1 - \sqrt{t_0/\tilde{t}}, \quad (1.37)$$

with $\tilde{t} = \tilde{t}_a$ and $z = E_b/E_a$. Neglecting the masses of b and c ,

$$t = z(1-z)\tilde{t}, \quad p_t^2 = z^2(1-z)^2\tilde{t}, \quad (1.38)$$

and the Sudakov form factor of Sec. 1.4.1 becomes

$$\tilde{\Delta}_i(\tilde{t}) \equiv \exp \left(- \sum_j \int_{4t_0}^{\tilde{t}} \frac{d\tilde{t}'}{\tilde{t}'} \int_{\sqrt{t_0/\tilde{t}'}}^{1-\sqrt{t_0/\tilde{t}'}} dz \frac{\alpha_s(z^2(1-z)^2\tilde{t}')}{2\pi} \hat{P}_{ji}(z) \right) \quad (1.39)$$

At large $\tilde{t}$ this falls more slowly than its collinear counterpart, though still faster than any negative power of $\tilde{t}$. The slower fall implies less branching, which is the suppression of soft gluon emission by angular ordering.

Chapter 2

Solenoidal Tracker At Relativistic Heavy Ion Collider

2.1 Relativistic Heavy-Ion Collider (RHIC)

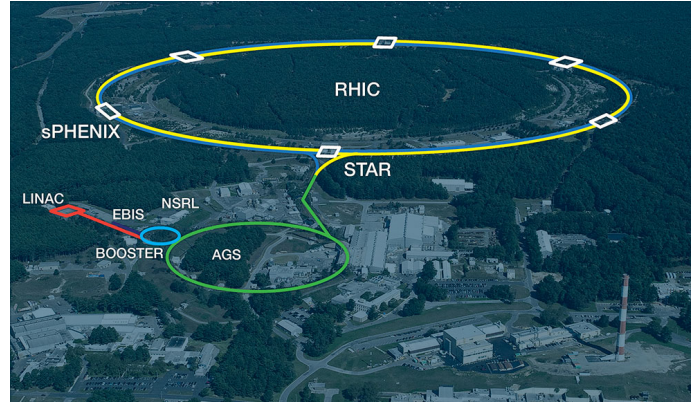


Figure 2.1: An aerial view of the Relativistic Heavy Ion Collider (RHIC), showing the locations of the facilities used for accelerating different ions, and the positions of the STAR and sPHENIX detectors.

The Relativistic Heavy Ion Collider (RHIC), at Brookhaven National Laboratory (BNL), is the only collider with dedicated running for heavy ion research (the CERN Large Hadron Collider runs heavy ions for only about one month per year), and the only polarized proton collider ever built. Two concentric accelerator rings 2.4 miles in circumference containing a total of 1,740 superconducting magnets afford RHIC the capability to independently accelerate and collide ion beam species from ${}^1_1\text{H}(p)$ to ${}^{238}_{92}\text{U}$ at energies between 3.85 GeV and 254.9 GeV per nucleon and different spin polarizations for protons (p), as shown in Fig. 2.1. These capabilities make RHIC the most flexible and capable collider in the world for transformative studies of extreme states of nuclear matter and the origin of the proton spin.

RHIC energies, species combinations and luminosities (Run-1 to 25)

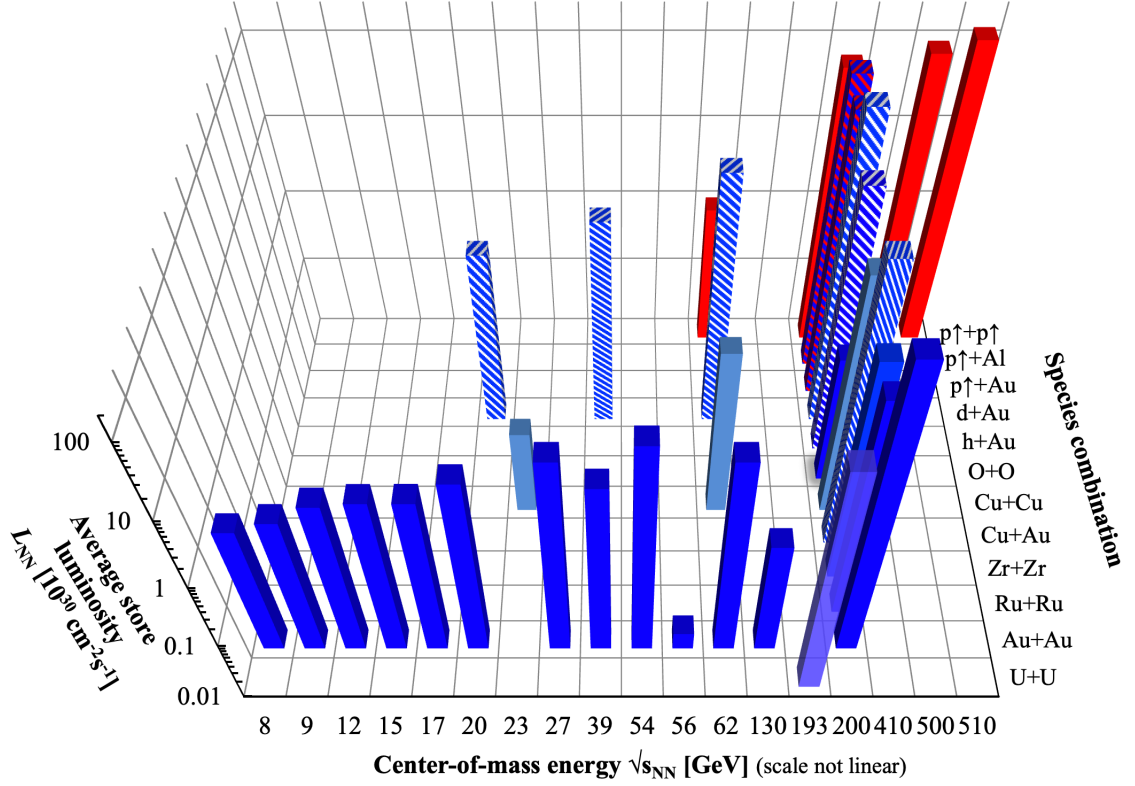


Figure 2.2: Summary of RHIC energies, species combinations, and luminosities for runs between 2000 and 2026. Taken from [?].

On February 6th 2026 RHIC ended operations with collisions of ^{16}O nuclei at $\sqrt{s_{NN}} = 200 \text{ GeV}$ recorded by the *Solenoidal Tracker At RHIC (STAR)* and *sPHENIX* detectors. Over its 25 years of active existence, it has performed ~ 300 trillion collisions of 49 unique pairs of nuclei, collected ~ 450 petabytes of data, whose analysis has led to over 650 publications. Eventually, one of RHIC's ion storage rings will be removed and replaced with a new ring for storing accelerated electrons inside the existing accelerator tunnel, and the other major components of the RHIC accelerator complex will live on as parts of the upcoming Electron-Ion Collider (EIC).

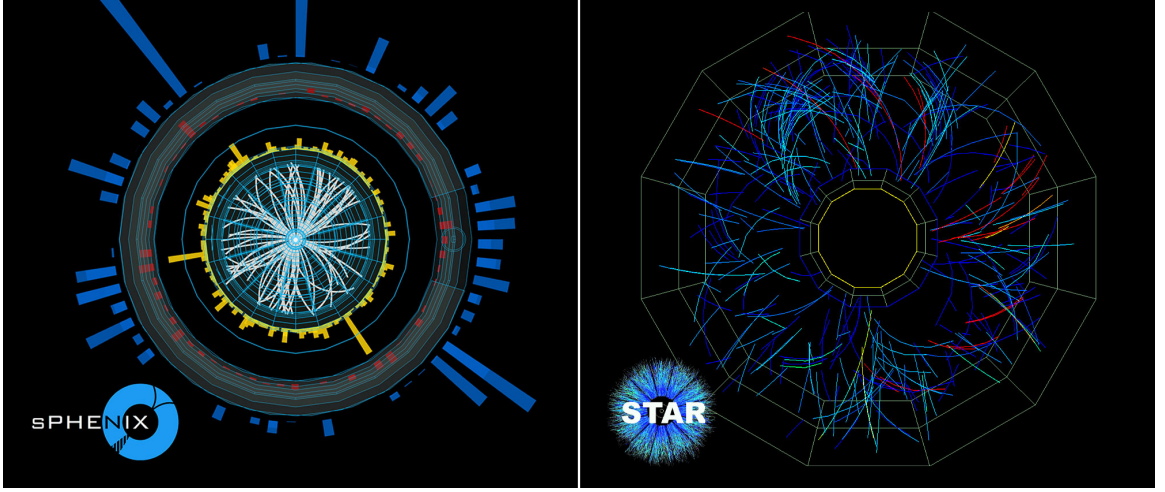


Figure 2.3: Some of the final O+O collisions at $\sqrt{s_{NN}} = 200$ GeV captured by the sPHENIX and STAR detectors at the RHIC.

2.1.1 From stable matter to little bangs

Before collisions, the nuclei have to be created first by taking the corresponding atoms and stripping off all their electrons. That process looks different for different nuclei.

- **For protons** it is *Optically Pumped Polarized H^- Ion Source (OPPIS)* fed into the *Linear Accelerator (LINAC)*. The LINAC accelerates them to ~ 200 MeV and injects into the *Booster Synchrotron* at which point, the two electrons are stripped away by a carbon foil. They are accelerated to upto 2.5 GeV and fed into the *Alternating Gradient Synchrotron (AGS)* which accelerates the proton beam to about 25 GeV.
- **For heavy-ions** like $^{197}_{79}\text{Au}$, a *Laser Ion Source (LION)* extracts Au^+ ions using laser ablation, which go into *Electron Beam Ion Source (EBIS)* which ionizes them to Au^{32+} and accelerates them to 2 MeV per nucleon using electron beams. Then they enter the booster synchrotron which accelerates them to about 95 MeV per nucleon. While exiting from the synchrotron, the atoms are stripped of more electrons by passing them through a stripping foil, and atoms in the Au^{77+} state are fed to the AGS. The beam's kinetic energy up to 8.86 GeV per nucleon for these Au atoms and the final two K shell electrons are stripped.

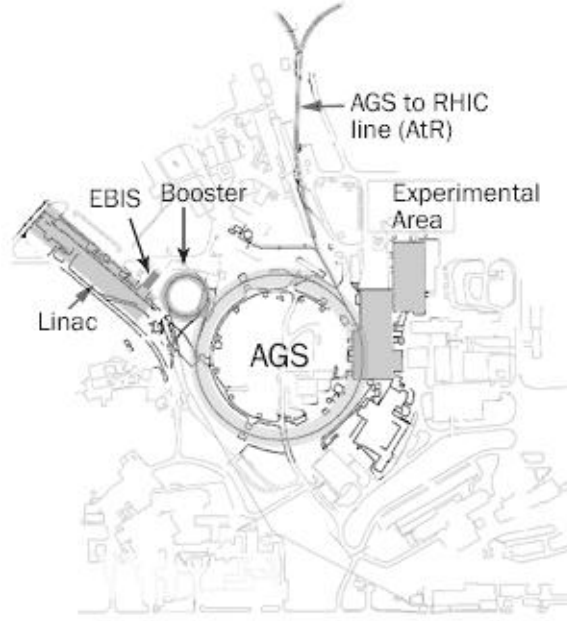


Figure 2.4: Schematic view of the AGS complex, showing various stages the ions go through before going into RHIC

when the ion beam reached top speed in the AGS, it was taken down another beamline called the AGS-to-RHIC (AtR) transfer line. At the end of this line, a switching magnet sent the ion bunches down each of two beamlines to fill RHIC's two ion storage rings. Bunches directed left filled the clockwise RHIC ring (blue) while those directed right traveled counterclockwise in the second RHIC ring (yellow). In the rings, superconducting radio-frequency (RF) cavities accelerate the ion bunches to energies of 100 GeV per nucleon for heavy ions (255 GeV for protons), with four cavities operating at 28.15 MHz to generate standing waves of electric fields for the acceleration and ten 197 MHz cavities maintaining beam stability during storage. Superconducting dipole magnets with 3.5 T fields keep the beams on a circular path, while quadrupole magnets focus the ion beams. A total of 1,740 superconducting magnets are used in the two rings at RHIC, cooled by liquid Helium at < 4.6 K. An aerial view of the RHIC complex with the different subsystems is shown in Fig 2.1.

There are 6 interaction regions (IR) around the RHIC complex, where the two beams cross and allow for detectors to be placed. 4 such detectors have existed during RHIC's

operation: At 10 o'clock and 2 o'clock, the PHOBOS and BRAHMS experiments operated only for the first few years of RHIC (ending in 2006). At 8 o'clock, the PHENIX experiment collected data from 2000 to 2016 and was then transformed into the sPHENIX detector which began operations in 2023. STAR sits at the 6 o'clock position, and has been the only experiment running for all 25 years of RHIC's existence.

2.2 The STAR detector

STAR is designed to provide hermetic coverage around the interaction point, emphasizing precise tracking capabilities in high multiplicity environments, for its main physics mission of obtaining a fundamental understanding of the microscopic structure of QCD interactions at high energy densities.

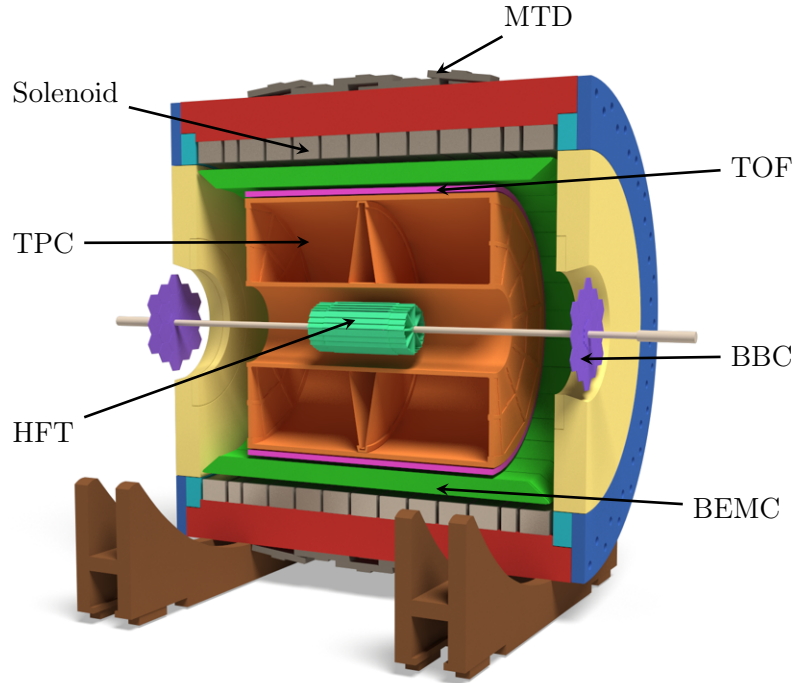


Figure 2.5: Schematic of the STAR detector, depicting the various detector subsystems.

It is a highly versatile, multipurpose detector with several subsystems used for triggering, tracking and identifying the particle species for charged tracks, and calorimetry. These subsystems are surrounded by a room-temperature solenoid magnet 6.2 m long and 5.26 m

in diameter which generates a magnetic field of 0.5 T in the detector volume. The detector is cylindrical with full coverage in the azimuthal direction in the mid-rapidity region along with wide coverage in pseudorapidity. A schematic diagram of the STAR detector with its various detector subsystems is shown in Fig. 2.5. As the only experiment running continuously since the inception of RHIC, STAR has undergone major upgrades that have augmented its tracking capabilities and increased its acceptance in the forward rapidity regions. The main detectors used for this analysis are the Time Projection Chamber (TPC) and Barrel Electromagnetic Calorimeters (BEMC).

2.2.1 Detector Coordinate System

Before we get to the components of the STAR detector system, we will introduce the coordinate system used for the calculations involved. STAR is cylindrical around the beamline, so a convenient choice for z -axis is the beamline going east-west. $z = 0$ is set at the *Interaction Point* (IP), center of the detector volume, as shown in Fig. 2.6

Kinematics of ultra-relativistic particles are described by its four-momentum, (E, p_x, p_y, p_z) . However, these coordinates are not additive under Lorentz boosts in the z -direction, making calculations involving relativistic entities somewhat unwieldy. Since p_x, p_y are already invariant under Lorentz boosts along the z -axis, we can just convert them to the convenient cylindrical coordinates

$$p_T = \sqrt{p_x^2 + p_y^2}, \quad \varphi = \arctan\left(\frac{p_y}{p_x}\right) \quad (2.1)$$

Energy (E) can be replaced with the particle's mass (m), or transverse mass $m_T = \sqrt{p_T^2 + m^2}$. We can make the z -coordinate additive under Lorentz Boosts along z -axis by representing a particle's direction along the z -axis by *rapidity*

$$y = \frac{1}{2} \ln \frac{E + p_z}{E - p_z} \quad (2.2)$$

However since y is mass-dependent and most detectors don't provide robust particle

identification at all kinematic ranges. We usually define the z -direction in a detector using *Pseudorapidity*, which is the massless limit of rapidity

$$\eta = \lim_{m \rightarrow 0} y = \frac{1}{2} \ln \frac{p + p_z}{p - p_z} = -\ln \left(\tan \frac{\theta}{2} \right) \quad (2.3)$$

Where θ is the polar angle between the particle's direction and the z -axis as shown in Fig. 2.6. Thus a particle's four-momentum is given by $(m_T, p_T, \eta, \varphi)$ or (p_T, η, φ, m) . See Appendix A.2 for a more detailed derivation of coordinates.

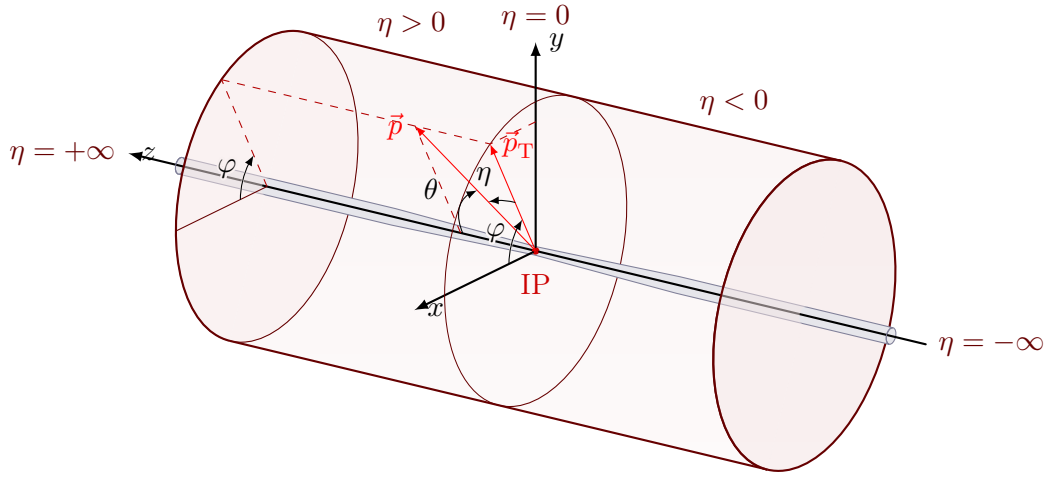


Figure 2.6: The (p_T, η, φ) coordinate system, drawn for a cylindrical collider detector. The beam axis defines z and the interaction point (IP) sits at the origin. A particle of momentum $\vec{p}$ is described by its transverse momentum $\vec{p}_T$, its azimuthal angle φ in the transverse plane, and its pseudo-rapidity η , fixed by the polar angle θ with respect to the beam.

2.2.2 TPC

Time Projection Chamber (TPC) **CITE!** was the the primary tracking detector for STAR, providing particle identification and momentum information for charged tracks by measuring their ionization energy loss (dE/dx) inside the TPC volume **CITE!**. As shown in Fig. 2.7 the TPC was a cylindrical detector 420 cm long with an outer (inner) radii of 200 (50) cm, concentric to the 0.5 T solenoid. Its acceptance covered a pseudorapidity range of $|\eta| < 1.0$, along with full azimuthal coverage. Particle momenta could be measured accurately over a

range of 0.1 to 30 GeV/ c , and particle identification was limited between 0.1 and 1 GeV/ c . The TPC volume contained an ionizable gas mixture called P10 (10 % methane, 90 % argon),

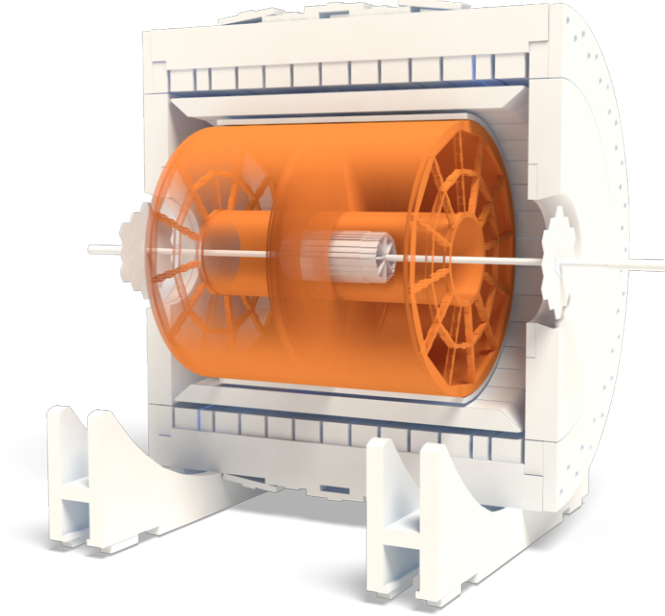


Figure 2.7: The Time Projection Chamber at STAR. Taken from [?].

chosen for its high drift velocity of 5.45 cm/ μ s in the well-defined, uniform electric field of ≈ 135 V/cm at STAR. The uniform electric field is defined between the thin conductive central membrane at $z = 0$ maintained at 28 kV acting as a cathode, and the two end caps each 210 cm away from the center at 0 V acting as anodes. The drift velocity can change over time based on atmospheric pressure fluctuations, therefore, calibrations are typically done using lasers every few hours during data-taking. To ensure minimal contamination by impurities like oxygen and water vapor maintained at 2 mbar above atmospheric pressure.

The charged particles leave ionization trails were captured by Multi-Wire Proportional Counters (MWPCs) **CITE!** with a segmented pad plane used in TPC readout. As particles ionize the TPC gas, freed electrons drift toward the MWPCs under the effect of the electric field. Signal contamination was prevented using a gated grid that initially restricted electron flow, opening only if a collision is deemed significant, allowing electrons to enter the MWPCs. Positive ions generated inside the MWPC are prevented from entering the TPC's drift

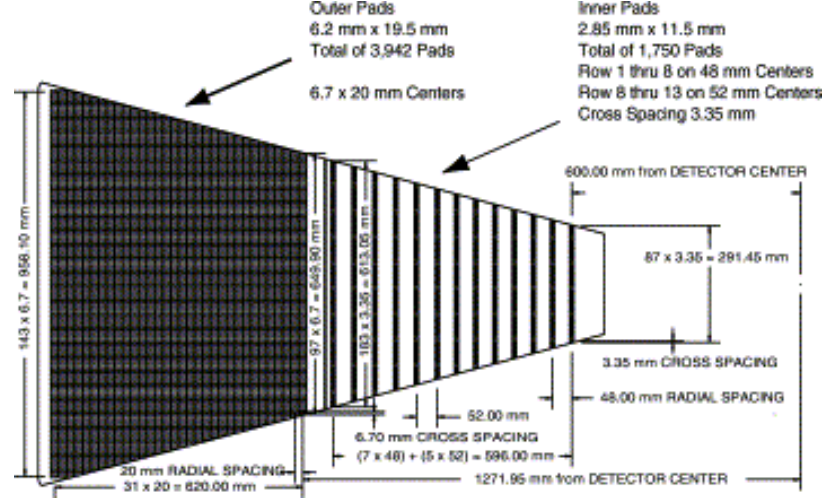


Figure 2.8: One of the twelve sectors of the TPC. Taken from [?].

volume using shield wires. The electrons passed the shield wires and were rapidly accelerated near the anode wires, creating avalanches through secondary ionization and amplifying the signal by a factor of 1,000 to 3,000. These avalanches induce signals in the surrounding pads without direct contact with the electrons. The MWPC pads detect the x-y position based on pad geometry — 3,942 pads across 32 rows in the outer sector and 1,750 pads in 13 rows in the inner sector. One such anode sector is shown in Fig. 2.8. The z information is then derived using the drift time and calibrated drift velocity of the electrons inside the detector volume. The typical resolution for position measurements from TPC is roughly 1 cm.

For a charged particle track, maximum of 45 hits could be recorded, which were fitted to a helix to calculate the particle momentum. Particles with momentum below 100 MeV/c curve too tightly in the magnetic field and do not reach the TPC, while those above 30 GeV/c follow nearly straight trajectories that compromise the measurement precision. For particles within these bounds, the relative momentum resolution is about 2 %. The direction of curvature gives the charge of the track. Energy loss as a function of distance travelled, dE/dx for a track was extracted using energy loss measured by a series of pads in the TPC. Figure 2.9 shows a typical plot of measured dE/dx as a function of momentum. For a given particle species with known momenta, the dE/dx curves are given by the Bichsel function

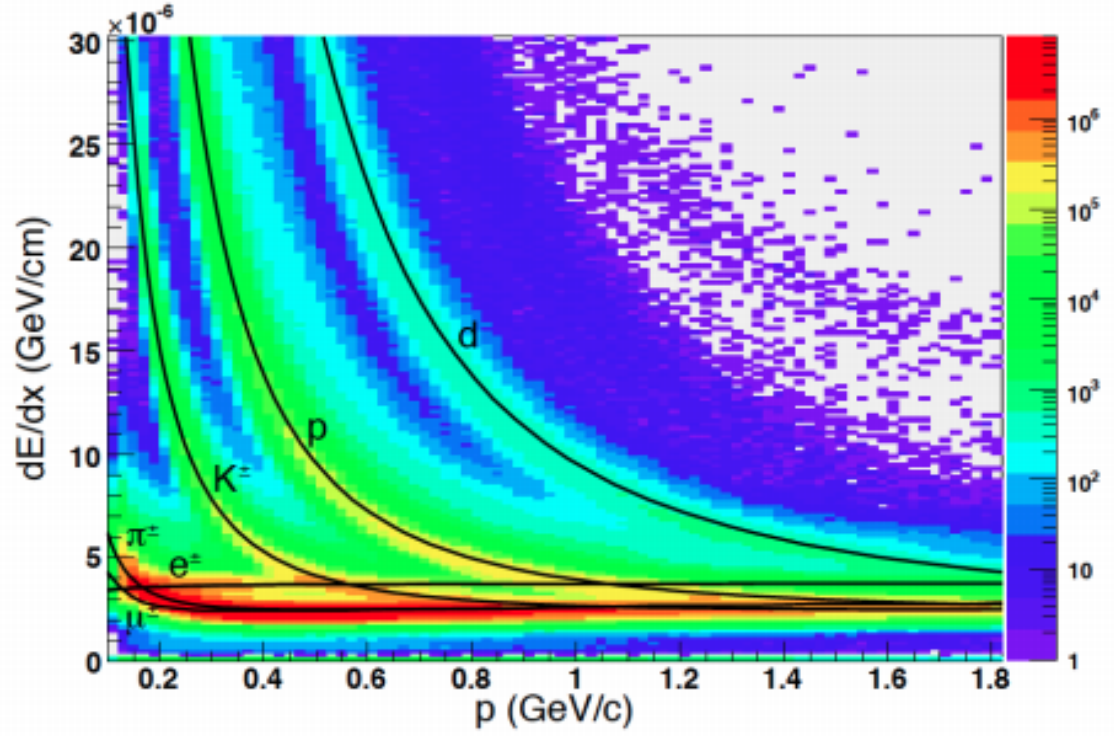


Figure 2.9: The ionisation energy loss for charged particles from TPC as a function of momenta. The letters represent different particles: $\mu^\pm$ for muons, $e^\pm$ for electrons, $\pi^\pm$ for pions, $K^\pm$ for Kaons, p for protons, and d for deuterons. The solid lines are the expected value of dE/dx for the corresponding particle species from the Bichsel function [?].

CITE! which are shown as solid lines in Fig. 2.9. For each particle species, the band has non-zero thickness due to the finite momentum resolution, and selection criteria based on simulations are used to assign particle ID. The particle identification from TPC is limited to low momenta tracks (about 1 GeV/c) only beyond which the bands for different species start to overlap.

2.2.3 BEMC

The STAR Barrel Electromagnetic Calorimeter (BEMC) [4] designed to measure particle energies, particularly for electron and neutral particle identification. BEMC was a fast detector, used as a trigger to select events containing a hard process. It also provided the neutral component of the jet energy that couldn't be reconstructed by the TPC in jet-related measurements. As shown in Fig. 2.10, it was cylindrical detector located between the TPC and magnet coils. It covers the pseudorapidity range $|\eta| < 1$ and the full azimuth range ($0 \leq \phi \leq 2\pi$), matching the TPC's coverage.

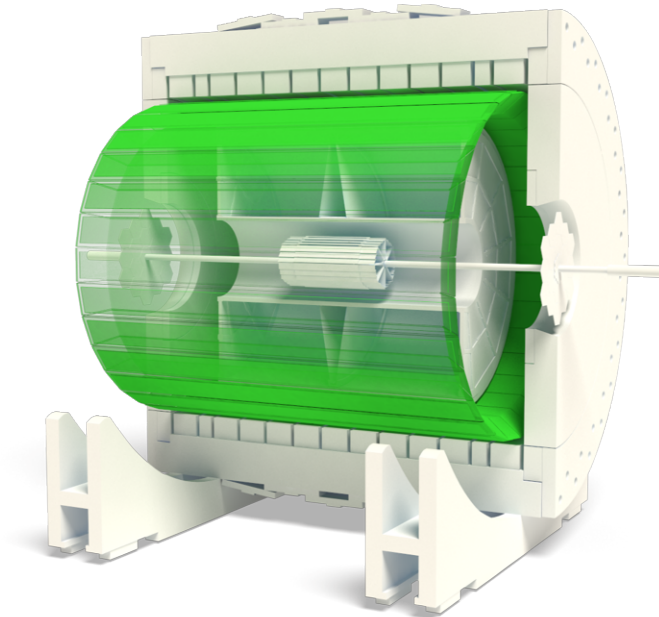


Figure 2.10: The Barrel Electromagnetic Calorimeter at STAR. Taken from [?].

The BEMC had inner and outer radii of 223.5 cm and 263 cm respectively and consisted

of 120 modules, each containing 40 towers, totaling 4800 towers. Each of the towers covers a range of $\Delta\eta \times \Delta\phi = 0.05 \times 0.05$, approximately $10 \text{ cm} \times 10 \text{ cm}$ near $\eta = 0$, growing slightly with increasing $|\eta|$ due to their projective geometry toward the center of the detector ($z = 0$). Each tower was built with 20 alternating layers of 5-mm thick lead, acting as absorbers to initiate electromagnetic showers, and 21 layers of plastic scintillators, which detect the photons produced by these showers. Together, the layers created a depth of around 20 radiation lengths (X_0), allowing the calorimeter to capture nearly all of the energy from electromagnetic showers. This depth is enough to contain 95 % of the energy from electrons and photons above 1 GeV. Within these layers, the shower's energy was most concentrated around 5-7 radiation lengths in, where the Shower Maximum Detector (SMD), a gas (90 % Argon, 10 % CO_2)-filled proportional counter improved spatial resolution to help differentiate between electromagnetic and hadronic shower profiles. The BEMC has an energy resolution that improves with particle energy, around 17% for 1.5 GeV electrons and 10% for 3 GeV electrons in head-on Au + Au collisions. This follows a typical calorimetric trend where energy resolution decreases as energy increases ($\sigma_E/E \approx 1.5\% + 15\%/\sqrt{E}$), providing complementarity with TPC where the resolution increases with increasing momentum ($\sigma_{p_T}/p_T \propto p_T$).

Chapter 3

Relativistic collisions of nuclei

Outgoing partons from hard processes undergo many successive splitting and fragmentation processes, becoming a collimated “spray” of hadrons called *jets* which are detected as dense clustering of energy in calorimeters. In principle, a *jet finding algorithm* should be able to faithfully get the kinematics of the initiating parton given the theory discussed in sec.1.4, which show partons cleanly branching into partons and eventually fragmenting into final-state hadrons that can be traced back to the shower partons. However reality is rarely this straightforward. First, the fragmentation functions that connect the partons and final state hadrons cannot be calculated in experiments. Second, it is not trivial to establish connection between final state particles and initial state partons even at the parton level, before we have the complications from hadronization and detector level reconstruction. For example, in a clean reaction like $e^+e^- \rightarrow q\bar{q}$. If one of the quark emits a gluon, then depending on the emission angle of the gluon, we might cluster 2 or 3 jets in the final state. We will discuss more about jet finding in 3.1

As discussed in Sec. 1.2.2, collisions of heavy-ions at near-light speeds can create the conditions for forming QGP. Sec. 2.1 discussed RHIC and sec. 2.2 describe the STAR detector system. Before moving on to analyses of heavy-ion collisions, we need to establish a few more concepts. Nuclei like $^{197}_{79}\text{Au}$, $^{96}_{40}\text{Zr}$, $^{96}_{44}\text{Ru}$, etc. are complex arrangements of protons and neutrons that have a multitude of non-trivial initial configurations leading up to their collisions which have significant effects on the final state particle distributions. Therefore, any measurement done using heavy-ion events need to report these initial configurations of the colliding nuclei. We discuss these in sec. 3.2.

3.1 Jets

To address the prevalent ambiguity of jet definitions, the Snowmass Accord **CITE!** stated that, experimental and theoretical measurements had to use the same jet finding algorithm on final state hadrons in order to be comparable. Thus clustered jets are not-quite the proxies for partons in the same way as parton showers, but regions in the phase space that are most likely to have contributions from parton showers. Even after we define a jet, another ambiguity comes from the decision of what constituents in the event must be used as inputs into the jet finding algorithm. Especially, in heavy-ion collisions jet-medium interactions make that choice non-trivial. Keeping these considerations in mind, a “good” jet finding algorithm is:

- *Collinear safe*, meaning it would find the same jet with the same $p_{T,\text{jet}}$ regardless of fragmentation details
- *Infrared safe*, meaning it would find the same jets, even in the presence of a large number of soft partons

The most commonly used family of infrared and collinear safe jet algorithms, sequentially recombines constituents in an event according to the following prescription:

1. For every pair of particles $\{i, j\}$ with momenta $(\mathbf{p}_i, \mathbf{p}_j) \equiv \{(p_{T,i}, \eta_i, \phi_i), (p_{T,j}, \eta_j, \phi_j)\}$

$$d_{ij} = \min(p_{T,i}^{2p}, p_{T,j}^{2p}) \frac{\Delta R_{ij}^2}{R^2}, \text{ and } d_i = p_{T,i}^{2p} \quad (3.1)$$

where, $\Delta R_{ij}^2 = (\eta_i - \eta_j)^2 + (\phi_i - \phi_j)^2$ is the (η, ϕ) distance between particle i and j . R is the resolution parameter of the algorithm, which decides the angular extent of clustered jets.

2. Find the minimum of the d_{ij} and d_i . If this minimum is a d_{ij} , combine these particles into one jet candidate, adding their energies and momenta, and return to the first step.

3. If the minimum is a d_i , this is a final state jet candidate. Remove it from the list and return to the first step. Iterate until no particles remain.

Above algorithm becomes the k_T *algorithm* for $p = 1$, *Cambridge-Aachen* for $p = 0$ and the most widely used *anti- k_T algorithm* for $p = -1$.

3.1.1 SoftDrop

The soft drop grooming procedure can be summarized as follows. First, the observed jet is reclustered with the Cambridge/Aachen (C/A) algorithm. Second, the soft drop criterion from Eq. 3.2 is checked at each clustering step when going backward through the C/A clustering tree.

$$\frac{\min(p_{T,1}, p_{T,2})}{p_{T,1} + p_{T,2}} > z_{\text{cut}} \left(\frac{\Delta R_{12}}{R} \right)^\beta \quad (3.2)$$

Here $p_{T,\{1,2\}}$ are the momenta of the two branches that had been merged together and ΔR_{12} is their geometric distance in the (η, ϕ) plane. When the softer branch fails the criterion it is removed from the jet and the procedure continues until it is satisfied. The remaining particles in the jet constitute the soft drop groomed jet. A convenient choice can be made for the soft threshold parameter z_{cut} and the angular exponent β . For $\beta = 0$ the soft drop procedure reduces to the modified mass drop tagger (mMDT)

3.1.2 Background Subtraction

Area based

Constituent based

3.2 Heavy-Ion Collisions

3.2.1 Initial State

3.2.2 Centrality Determination

In heavy-ion collisions, the impact parameter (b) of collisions is defined by the distance between the geometrical centers of the colliding nuclei in the plane transverse to their direction.

A schematic diagram of the geometry of heavy-ion collisions is shown in Fig. 3.1.

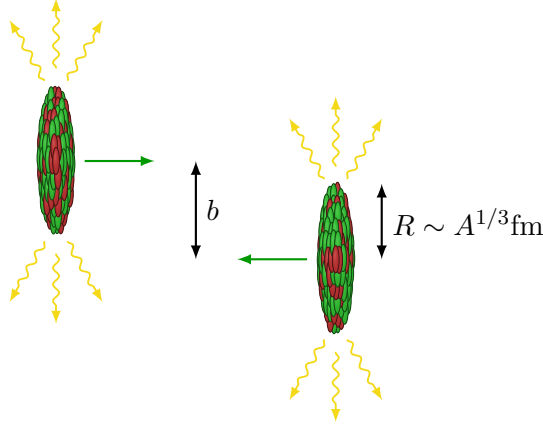


Figure 3.1: Typical heavy-ion collision

As the impact parameter is not measurable in experiments, the charged particle multiplicity of events can be used to define collision centrality as they reflect the initial geometry of the colliding nuclei. To obtain the geometrical quantities like the impact parameter, the number of participant nucleons (N_{part}) and the number of binary collisions (N_{coll}), one has to rely on model calculations which involves mapping the charged particle multiplicity from experiment to that of a simulated one. The widely used model for determining these quantities is the Glauber Monte Carlo (Glauber MC) model. In Glauber model, nucleons

in a nuclei are distributed following the Wood–Saxon density distribution. The nuclei are then assigned a random impact parameter and translated to collide. The distance between the centers of nucleons of each nucleus is calculated. If this distance is found to be less than $\sqrt{\sigma_{\text{inel}}^{\text{NN}}/\pi}$, where $\sigma_{\text{inel}}^{\text{NN}}$ is the inelastic nucleon–nucleon cross section, the nucleons have undergone a binary collision. Nucleons that have undergone at least one binary collision are said to have participated in the reaction. This procedure is followed to obtain N_{part} , N_{coll} event-by-event.

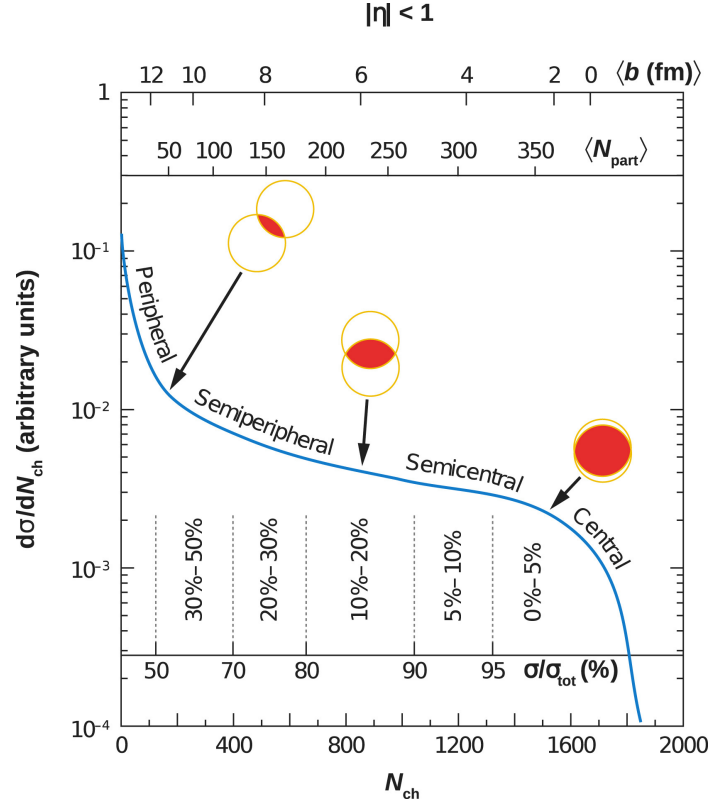


Figure 3.2: An illustration of the total inclusive charged particles multiplicity N_{ch} and number of participants (N_{part}) and impact parameter (b) in heavy-ion collisions

A class of events belonging to a given centrality (say 0%–5%) is obtained by the collection of events representing the fraction of the total cross-section. Glauber model can only provide the information on N_{part} , N_{coll} and impact parameter b , but for mapping with the experimental data, one has to use a combination of the Glauber model and a particle production model called the Two-Component Model. Assuming that particle production is

governed by contribution from hard component (proportional to N_{coll}) and soft component (proportional to N_{part}), it simulates the multiplicity distribution according to

$$N_{\text{ch}} = n_{pp} \left[x N_{\text{coll}} + (1 - x) \frac{N_{\text{part}}}{2} \right] \quad (3.3)$$

where the quantity n_{pp} is the multiplicity from a $p + p$ collision of the same center-of-mass energy. x is the contribution of hard component. n_{pp} is drawn from a negative binomial distribution with parameters NBD $(n_{pp}; \langle n_{pp} \rangle, k)$, where $\langle n_{pp} \rangle$ is the average multiplicity in the $p + p$ collision and k is the width of the distribution. The simulated charged multiplicity distribution is then fitted with the corresponding multiplicity distribution from the experiment. $\langle n_{pp} \rangle$, k , and x are kept as free parameters in the fitting to experimentally measured charged particle multiplicity distribution. Initial guess values for x could be set close to those determined in different experiments for faster convergence of fitting.

Fig. 3.2.2 illustrates collision centrality definition in heavy-ion collisions by comparing the charge particle multiplicities with Glauber MC + Two-component model simulation. Centrality classes are classified for the simulated distribution. $\langle N_{\text{part}} \rangle$, $\langle N_{\text{coll}} \rangle$ in those centrality classes are calculated.

3.2.3 Event plane reconstruction

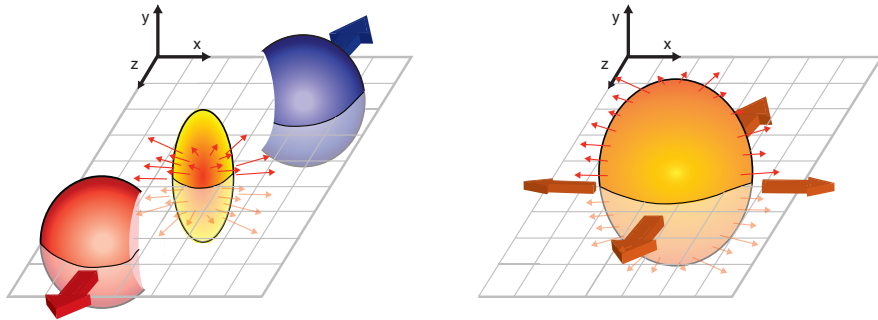


Figure 3.3: Schematic diagrams showing the initial overlap region (left) and the spatial anisotropy generated by this anisotropic overlap region. This anisotropy can be quantified using the Fourier coefficients of the momentum anisotropy.

Within the overlap region, symmetry planes are generated from initial asymmetries in the nucleon distributions and can be quantified by a harmonic decomposition [5]. The reaction plane would correspond to the second-order symmetry plane $\Psi_{2,EP}$ if nucleon distributions were in their average positions and devoid of fluctuations of interactions amongst nucleons [6]. We refer in this letter to the event plane as being the experimentally reconstructed second-order symmetry plane [5].

By measuring the charged particle azimuthal distribution, the n -th order event plane can be extracted by [5]:

$$\Psi_{n,EP} = \frac{1}{n} \arctan \left(\frac{Q_{y,n}}{Q_{x,n}} \right), \quad (3.4)$$

Where, the weighted Q vectors are given by:

$$Q_{x,n} = \sum_{\text{tracks}} w_{\text{track}} \cos(n\phi_{\text{track}}) \quad \& \quad Q_{y,n} = \sum_{\text{tracks}} w_{\text{track}} \sin(n\phi_{\text{track}}). \quad (3.5)$$

where the sum is calculated for all reconstructed charged particles (tracks) in the event, ϕ_{track} is the track's azimuthal angle, and w_{track} the weight associated with the track. Weights, w_{track} , are optimized to calculate the event-plane vector to the best accuracy. This work uses the common approach of scaling by the track's p_T ($w_{\text{track}} = p_{T,\text{track}}$) [5]. The event plane is calculated event-by-event, following a procedure similar to Ref. [7] using charged tracks with $0.2 < p_{T,\text{track}} < 1.0$ GeV/ c measured within the TPC. The approach is called the Modified Reaction Plane (MRP) method [2]. Additional details can be found in Refs. [7, 8].

The impact of highly energetic jets on the calculation of the event-plane orientation is reduced by removing the particles within the pseudorapidity strip ($|\Delta\eta| < 0.4$) across $\Delta\phi$ surrounding the leading jet. This procedure also removes a significant portion of the away-side jet, located opposite in azimuth. An upper limit of 1.0 GeV/ c is used in the calculation of the event plane to exclude the momentum range of particles used in correlation functions

from the calculation of the event plane which is used to characterize the near-side jets. Due to finite acceptance and multiplicities, the calculated event plane has an underlying anisotropy that is corrected by applying two separate correction methods.

First, a calibration and recentering correction procedures are applied to remove bias introduced by non-uniform acceptance of the TPC tracking system and further account for potential beam-condition effects[5, 9, 10]. This procedure involves recentering the weighted Q -vectors such that, $\langle Q_{x,n} \rangle = 0 = \langle Q_{y,n} \rangle$.

Recentering is done by calculating a modified Q -vector, obtained by subtracting an event averaged Q -vector from each event's nominal Q -vector and done for 10% centrality intervals and 4 cm v_z intervals. The recentering approach, which drastically improves the uniformity of the event plane, is however, unable to remove the higher harmonics of $\Psi_{n,EP}$ [5]. To help remove higher harmonics and make the event-plane angle isotropic in the lab frame [11], a second correction step, referred to as shifting, is applied event-by-event. This method defines a new angle:

$$\Psi'_{2,EP} = \Psi_{2,EP} + \sum_n \frac{2}{n} (-\langle \sin(n\Psi_{2,EP}) \rangle \cos(n\Psi_{2,EP}) + \langle \cos(n\Psi_{2,EP}) \sin(n\Psi_{2,EP}) \rangle), \quad (3.6)$$

where the brackets denote an average over events. We require the vanishing of each n -th Fourier moment up to 20th order. Similarly to recentering, the shifting correction is done separately for 10% centrality intervals and 4-cm v_z intervals. Additional details of the recentering and shifting corrections can be found in Refs. [5, 9, 12]

The resulting azimuthal anisotropy can be characterized by the Fourier decomposition of the azimuthal particle distribution with respect to the second-order event plane [13, 14]:

$$\frac{dN}{d(\phi - \Psi_{RP})} = \frac{N_0}{2\pi} \left(1 + 2 \sum_{n=1}^{\infty} v_n \cos[n(\phi - \Psi_{RP})] \right), \quad (3.7)$$

where N_0 is the number of particles, ϕ describes the azimuthal angle of the particles, Ψ_{RP}

describes the azimuthal angle of the true reaction plane determined by the beam axis and the impact parameter and v_n is the n -th harmonic (flow) coefficient. Ψ_{RP} is not experimentally known and is replaced by the reconstructed event-plane angle. Due to finite event multiplicity, there will be a difference between these two planes. It is quantified by event-plane resolution, R_n or $R_n\{\Psi_{2,EP}\}$ given by Eqn. 3.8. The observed v_n , v_n^{obs} is corrected for this limited resolution by doing, $v_n = v_n^{\text{obs}}/R_n$ [5, 15]. Because an ideal event-plane resolution is equal to 1, for non-ideal cases, the value of the coefficients will be raised by applying the correction. Thus, R_n impacts the flow-modulated background for these correlations, as described in Sect. 5.1.3,

$$R_n = R_n\{\Psi_{2,EP}\} = \langle \cos(n[\Psi_{RP} - \Psi_{2,EP}]) \rangle \quad n \text{ is even} \quad (3.8)$$

Furthermore, individual events are divided into two random sub-events by assigning charged tracks to sub-events, “ a ” and “ b ”. The sub-events are unique, with approximately equal multiplicities. We can write the correlation of two event planes by taking the product of two sub-events [5, 16]:

$$\langle \cos(n[\Psi_{2,EP}^a - \Psi_{2,EP}^b]) \rangle = \langle \cos(n[\Psi_{2,EP}^a - \Psi_{RP}]) \rangle \langle \cos(n[\Psi_{2,EP}^b - \Psi_{RP}]) \rangle, \quad (3.9)$$

This allows calculation of the event-plane resolution directly from data. Since a and b have equal multiplicities, the resolution of each sub-event can be calculated from the correlation between the two sub-events as [5]:

$$\langle \cos(n[\Psi_{2,EP}^a - \Psi_{RP}]) \rangle = \sqrt{\langle \cos(n[\Psi_{2,EP}^a - \Psi_{2,EP}^b]) \rangle}. \quad (3.10)$$

The event-plane resolution is multiplicity dependent and calculated for separate ranges of collision centrality using the two sub-events method. Narrower bins are calculated and then

combined accordingly to match the ranges used by this analysis, by averaging the results from the narrow bins weighted by the multiplicity of each bin [11].

3.3 Spacetime picture of QGP

Estimates of the energy density indicate that central heavy ion collisions with an incoming energy per nucleon pair as low as $\sqrt{s_{NN}} = 7.7$ GeV, the lower boundary of collision energies accessible at RHIC, can reach energy densities above 1 GeV/fm^3 **CITE!** and that collisions at 2.76 TeV, accessible at the LHC, reach energy densities as high as 12 GeV/fm^3 **CITE!**. Contrary to initial naïve expectations of a gas-like QGP, the QGP formed in these collisions was shown to behave like a liquid of quarks and gluons **CITE!**.

The heavy ion collision and the evolution of the fireball, as depicted in Figure **CITE!**, has several stages, and the measurement of the final state particles can be affected by one or all of these stages depending on the production mechanism and interaction time within the medium.

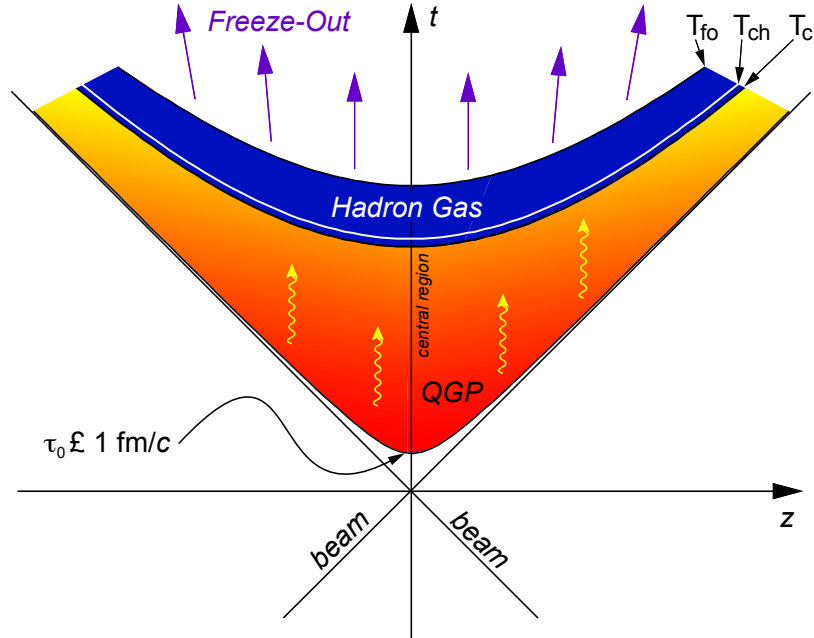


Figure 3.4: A light cone diagram showing the stages of a heavy ion collision. The abbreviation T_{fo} is for the thermal freeze-out temperature, T_{ch} is for the chemical freeze-out temperature, and T_c is for the critical temperature where the phase transition between a hadron gas and a QGP occurs. τ_0 is the formation time of the QGP.

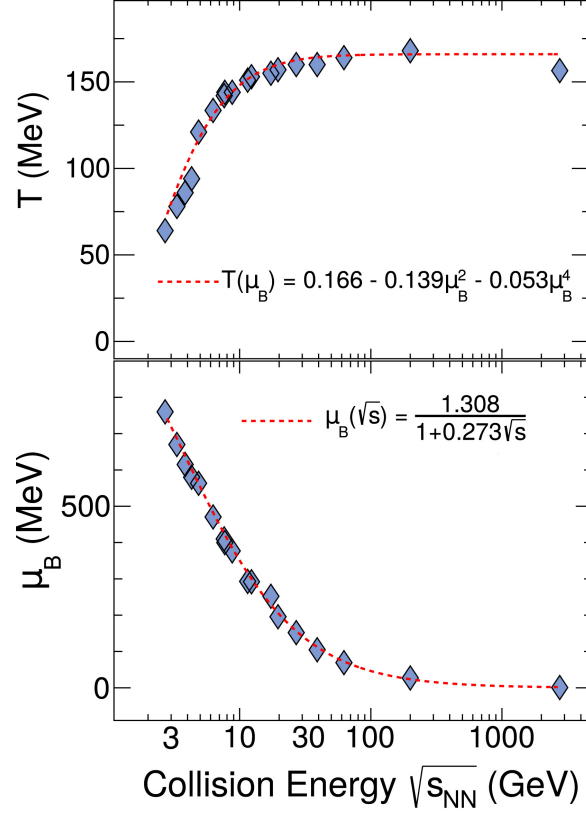


Figure 3.5: Chemical freeze-out parameters T and μ_B obtained by thermal model fits to experimental data on particle yields as a function of collision energies. Red line represents parametrizations of T and μ_B as a function of $\sqrt{s_{NN}}$

3.4 Signatures of QGP

Since

Since QGP is short-lived, the properties of the QGP medium can only be studied from the final state particles measured by the detectors. These "final state" particles carry imprints of the medium's properties which influence their final momenta distributions and correlations. Among the key observables in this regard is the azimuthal anisotropy in the momentum distribution of the final state particles, denoted as the *anisotropic flow*

Chapter 4

Data analysis techniques

As a result of the quantum-mechanical nature of the collisions and the interaction of their products with the detectors, the observations resulting from a particular interaction are fundamentally probabilistic. Therefore, the approach to data analysis and the conclusions drawn from the data must be framed in statistical terms, which include not only low-level tasks, such as the reconstruction and identification of particles and the measurement of their energies and momenta, but also high-level tasks, such as searches for new particles and measurements. The high dimensionality and large volume of LHC data pose a problem because the statistical model over the high-dimensional space of their experimental data is not known explicitly in terms of an equation that can be evaluated. Instead, one typically has access to large samples of simulated data generated by stochastic simulation programs that model the physics of particle interactions on various scales. If the data were fairly low dimensional ($d < 5$), the problem of estimating the unknown statistical model from the simulated samples would not be difficult. Histograms or kernel-based density estimates provide reasonable estimates in low-dimensional spaces. These strategies, however, suffer from the curse of dimensionality. In a single dimension, N samples may be required to describe the source probability density function. In d dimensions, the number of samples required increases to the power of the data's dimensionality: $\mathcal{O}(N^d)$. The consequence is that any dimensionality greater than five or ten requires impractical or impossible computational resources, regardless of the speed of the sample generator.

4.1 Machine Learning Basics

Fortunately, many of the tasks encountered in HEP can be naturally reformulated as machine learning problems. Typically, the problems are formulated in terms of a search for some function $f : X \rightarrow Y$, from the space of the observed data X to a low-dimensional space of a desired target label Y , which optimizes some metric of our choosing. This metric is called a loss function and is written as $L[\mathbf{y}, f(\mathbf{x})]$.

Ideally, a learning algorithm would find the function that optimizes L over all possible values of $(\mathbf{x}, \mathbf{y})$, but this is intractable owing to the curse of dimensionality and an infinite number of functions to choose from. Instead, in supervised learning, one has labeled training data $\{\mathbf{x}_i, \mathbf{y}_i\}_{i=1}^N$ sampled from $p(\mathbf{x}, \mathbf{y})$. Furthermore, the function space is restricted to a model—a highly flexible family of functions $f_\phi(\mathbf{x})$ parameterized by ϕ . In this case, the algorithms minimize the loss function directly with respect to the model parameters ϕ . Neural networks, support vector machines, and decision trees are examples of types of models commonly used in machine learning. These models often have a large number of parameters, and in the case of neural networks, finding the optimal f_ϕ can be a difficult problem

An essential goal in machine learning is generalization—the ability of the model to perform well on data that were not used in training. Failure in this task is called overtraining or overfitting. A vast array of techniques exist to avoid overtraining, all of which can be considered forms of regularization. Regularization techniques such as dropout were key to advances in image recognition with deep learning

4.2 Jet Classification

4.3 Unfolding

4.4 Uncertainties in ML predictions

Chapter 5

Jet-hadron correlations relative to the event plane

Historically, the effects of the medium on high energy partons were studied by measuring the leading particles and studying the p_T distribution of pairs of high- p_T particles. Leading particle studies, however, have limitations in extreme quenching scenarios. In these cases, particle emission occurs near the surface of the medium and thus the sensitivity of the leading particles to the region of highest energy density is very limited. This consequently makes the nuclear modification factor insensitive to medium properties. Full jet reconstruction allows for unbiased measurement of the original parton 4-momentum and the transverse and longitudinal shape of the jet. Through modifications of the jet structure, properties of the medium can be studied.

When studying the $\Delta\phi$ distributions, a peak is observed at $\Delta\phi = 0$ which is the triggered jet. This side is referred to as the near-side. There is another peak which is seen near π . This is referred to as the away-side. The away-side consists of the hadrons associated with the trigger jet and make up the second jet which results from the early hard-scatter.

In this analysis, the coordinate system used to depict the distribution of associated particles is defined relative to a reconstructed jet, also called a trigger jet. The distribution is thus given by:

$$\frac{1}{N_{\text{trig}}} \frac{d^2 N_{\text{assoc,jet}}}{d\Delta\phi d\Delta\eta}, \quad (5.1)$$

where N_{trig} is the number of trigger jets, $N_{\text{assoc,jet}}$ is the number of associated particles, $\Delta\phi$ ($= |\phi_{\text{jet}} - \phi_{\text{assoc}}|$) is the azimuthal angle of those trigger particles relative to the trigger

jets, and $\Delta\eta$ ($= |\eta_{\text{jet}} - \eta_{\text{assoc}}|$) is the difference in the pseudorapidities of the trigger jet and associated particle. The goal of this analysis is to study the conditional yield of associated particles, the width of the near- and away-side peaks (quantified using the gaussian width) as a function of the angle between the jet axis and the event plane. The yield is estimated by:

$$\text{Yield} = \frac{1}{N_{\text{trig}}} \int_c^d \int_a^b \frac{d^2 N_{\text{assoc,jet}}}{d\Delta\phi d\Delta\eta} d\Delta\phi d\Delta\eta. \quad (5.2)$$

The choice of integration limits is somewhat arbitrary. They are chosen based on practical considerations, including the detector acceptance and binning of histograms.

5.1 Analysis Method

This measurement utilizes data collected during the 2014 run from Au+Au collisions at nucleon-nucleon center-of-mass energy of $\sqrt{s_{\text{NN}}} = 200$ GeV by the STAR experiment [17] at RHIC. Events referred to as signal (same) events are required to contain a HT trigger in the BEMC [18]. Minimum-bias (MB) triggered events based on coincidence of Zero Degree Calorimeters (ZDC coincidence), Vertex Position Detectors and Beam-Beam Counters signals are used to estimate the pair-acceptance effects via a mixed-event (ME) technique [19]. For this analysis, 9.4M HT-triggered and 4.0M MB collision events are used. Events are further categorized by their centrality selection, defined in section 3.2.2. The events are required to have a reconstructed primary vertex $|v_z| < 24$ cm and centrality of 20-50%.

The reaction plane is approximated by the second-order event plane, which is the experimentally reconstructed second-order symmetry plane, and will be referred to as the “event plane” ($\Psi_{2,EP}$) in this text, for simplicity. The distributions of these associated tracks relative to the trigger jet are measured in three bins in the angle between the trigger jet and the event plane, in-plane ($|\Psi_{2,EP} - \phi_{\text{jet}}| < \pi/6$), mid-plane ($\pi/6 < |\Psi_{2,EP} - \phi_{\text{jet}}| < \pi/3$), and out-of-plane ($|\Psi_{2,EP} - \phi_{\text{jet}}| > \pi/3$) bins. The analysis is restricted to 20–50% central

Au+Au collisions to achieve the highest event-plane resolution and therefore the analysis will be most sensitive to any path-length dependencies.

The second- (fourth-) order event-plane resolutions relative to the second-order event plane ($R_2\{\Psi_{2,EP}\}$ and $R_4\{\Psi_{2,EP}\}$ respectively) as a function of collision centrality are shown in Fig. 5.1. The resolution is peaked around the 20-30% and 30-40% centrality. The event-plane resolutions for $R_2\{\Psi_{2,EP}\}$ and $R_4\{\Psi_{2,EP}\}$ were 0.56 and 0.28, respectively. The errors on the event-plane resolution calculation were less than 1%, leading to a negligible effect on the final $\Delta\phi$ correlations. Measured values for the event-plane resolution are in good agreement with prior STAR studies [2]. These resolutions are evaluated to correct the observed flow coefficients which arise in the fits of the combinatorial background discussed in Sect. 5.1.3.

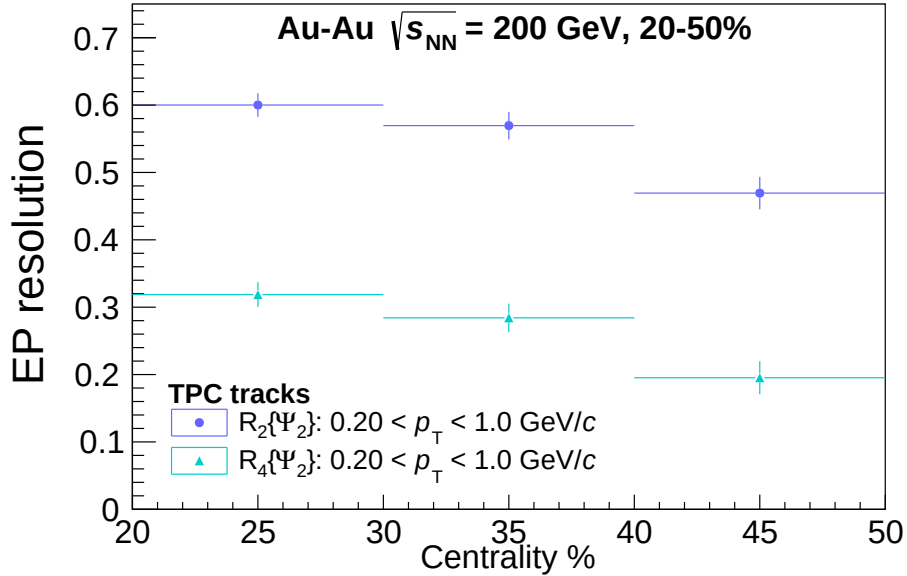


Figure 5.1: Event-plane resolution: Second-order (fourth-order) harmonic relative to the event plane, $R_2(\Psi_2)$ ($R_4(\Psi_2)$), respectively. The approach follows the Modified Reaction Plane (MRP) method [2] utilizing the charged tracks of the TPC for event-plane reconstruction and resolution calculation for tracks ranging from 0.2-1.0 GeV/c.

5.1.1 Jet reconstruction and selection

Full jets are reconstructed by measuring charged-tracks in the TPC and collecting neutral-particle information from the BEMC. The anti- k_T algorithm clusters these particles into jets by reconstructing the jet momenta as the quadratic sum of their constituent momenta using a boost-invariant p_T^2 recombination scheme. Tracks used for reconstructing jets are assumed to be pions while the towers to have arisen from massless particles. All jets measured in this work are clustered with resolution parameter of $r = 0.4$. The area of jets, A_{jet} , is found with FastJet using active ghost particles [20]. Partially reconstructed jets at the edge of the acceptance are rejected by applying a fiducial cut, $|\eta_{\text{jet}}| < 1.0 - r$, to assure all jets fall within the acceptance of the detectors.

Jets produced in heavy-ion collisions sit on top of a large amount of underlying event. The jet signal can be found beneath tens to hundreds of particles resulting from various other processes. To reduce the influence of these background particles, this analysis requires tracks (towers) with $p_T(E_T) > 2.0 \text{ GeV}/c$ for jet reconstruction. At RHIC energies this selection reduces the median background energy density per unit A_{jet} , $\langle \rho \rangle$, down to $\approx 0.6 \text{ GeV}$. This high-constituent selection, referred to as a “hard-core” jet selection reduces fluctuations, fake jets, and background jet particles [21]. To further reduce contributions from the background and to match the trigger condition, the jets are required to contain a constituent tower that fired the HT trigger ($E_{T,\text{tower}} > 5.4 \text{ GeV}/c$). The remaining jets are studied in classes of jets with $15 < p_{T,\text{jet}} < 20 \text{ GeV}/c$ and $20 < p_{T,\text{jet}} < 40 \text{ GeV}/c$.

5.1.2 Jet-hadron correlation

The measurement of the correlation function (distribution of charged hadrons relative to reconstructed jets) described in Eqn. 5.1 requires several corrections. The correlation function is measured in pseudorapidity ($\Delta\eta$) and azimuth ($\Delta\phi$) as:

$$\frac{1}{N_{\text{trig}}} \frac{d^2 N_{\text{assoc,jet}}}{d\Delta\phi d\Delta\eta} = \frac{1}{N_{\text{trig}}} \frac{1}{\epsilon(p_{\text{T,assoc}}, \eta_{\text{assoc}})} \frac{1}{a(p_{\text{T,assoc}}, \Delta\phi, \Delta\eta)} \left(\frac{d^2 N_{\text{assoc,jet}}^{\text{meas}}}{d\Delta\phi d\Delta\eta} - \frac{d^2 N_{\text{assoc,jet}}^{\text{bkgd}}}{d\Delta\phi d\Delta\eta} \right). \quad (5.3)$$

$N_{\text{assoc,jet}}$ gives the number of pairs of trigger jets and the associated hadrons, $N_{\text{assoc,jet}}^{\text{meas}}$ and $N_{\text{assoc,jet}}^{\text{bkgd}}$ are the number of pairs measured and the pair characterized as background respectively. $\epsilon(p_{\text{T,assoc}}, \eta_{\text{assoc}})$ is the single-track reconstruction efficiency¹. The pair acceptance, $a(p_{\text{T,assoc}}, \Delta\phi, \Delta\eta)$, is calculated from the raw pairs that we measure from a trigger jet associated with charged hadrons from mixed events.

The correlations are determined in bins of centrality, reconstructed trigger-jet transverse momentum ($p_{\text{T,jet}}$), associated-hadron transverse momentum ($p_{\text{T,assoc}}$), and bins of the trigger jet relative to the event plane (in, mid, out, all combined angles) defined in Sect. 5.1. The corrected correlation functions contain a large combinatorial background ($N_{\text{assoc,jet}}^{\text{bkgd}}$), which must be subtracted. This subtraction procedure is described in Sect. 5.1.3 .

The pair-acceptance correction, $1/a(p_{\text{T,assoc}}, \Delta\phi, \Delta\eta)$, accounts for the detector and tracking inefficiencies, and is found by correlating jets from HT-triggered events with associated hadrons from MB events of the same event class. In addition to providing the acceptance correction, the mixed-event procedure will also help remove the trivial correlation due to an η dependence in the single-particle track distributions [6]. Taking the ratio of the single-event pairs by the mixed-event pairs removes the detector effects of acceptance and efficiency. The pair acceptance will serve as the dominant effect, given that there is little η dependence in both the tracks and jets across the acceptance range of this analysis.

The event-mixing procedure used in this analysis is well described in Ref. [19]. The mixed events used for calculating $a(p_{\text{T,assoc}}, \Delta\phi, \Delta\eta)$ in this work are required to be within the same 10% centrality class and to have a vertex position within 4 cm along the direction of the beam (z). They are constructed separately for 20-30%, 30-40%, and 40-50% centrality classes

¹Acceptance effects such as gaps between sector boundaries are included in this correction, however, we refer to $\epsilon(p_{\text{T,assoc}}, \eta_{\text{assoc}})$ as an efficiency correction in order to distinguish it from $a(p_{\text{T,assoc}}, \Delta\phi, \Delta\eta)$.

and combined accordingly. High-momentum tracks are nearly straight, so the detector acceptance does not change significantly at high- $p_{T,\text{assoc}}$, and thus all associated momentum bins greater than 2.0 GeV/ c are combined to increase statistics. There is no difference in efficiency and acceptance within uncertainties for different orientations of the jet relative to the event plane, and therefore the same correction $a(p_{T,\text{assoc}}, \Delta\phi, \Delta\eta)$ is applied for all angles relative to the event plane. The acceptance $a(p_{T,\text{assoc}}, \Delta\phi, \Delta\eta)$ is normalized to 1 at its maximum, determined using the region of approximately constant acceptance ($|\Delta\eta| < 0.4$). For each $p_{T,\text{assoc}}$ bin, the projection of the flat plateau region (integrated over the region $|\Delta\eta| < 0.4$) was fit with a constant over the whole $\Delta\phi$ range. The associated uncertainties of the fits were used for the systematic uncertainty on the mixed-event normalization, which is added in quadrature and reported as the scale uncertainty. This systematic uncertainty is under 1% (1.25%) in all $p_{T,\text{assoc}}$ bins used for the reported results of jets with $15 < p_{T,\text{jet}} < 20$ GeV/ c and $20 < p_{T,\text{jet}} < 40$ GeV/ c .

A shape uncertainty in $\Delta\phi$ due to the z_{vtx} binning, similar to that in $\Delta\eta$, could lead to an additional shape uncertainty in the correlation functions. To test for this, the ratio of the 1D $\Delta\phi$ projection with the nominal z_{vtx} binning and the binning described above was calculated for each $p_{T,\text{jet}}$, $p_{T,\text{assoc}}$, and centrality bin. The variations about 1.0 are smaller than the statistical errors associated with the points. This uncertainty was therefore considered negligible.

There is an additional shape uncertainty associated with the application of the acceptance correction due to slight changes in the correlation function at large $\Delta\eta$ in the acceptance with v_z position. The background level is determined from the level of the correlation function at large $\Delta\eta$, leading to a scale uncertainty in the background subtraction dependent on $p_{T,\text{assoc}}$. This uncertainty is from the differences between the nominal (z -integrated) and the z -vertex method (z -binned) for correcting the mixed events on the level of the background in the $0.6 < \Delta\eta < 1.2$ range, and signal plus background, in the $|\Delta\eta| < 0.6$ range. The large $\Delta\eta$ region is used to determine the background so any uncertainties in the level of the correlation

function in this region lead to an uncertainty in the level of the background in the signal region. This is expressed as an additional scale band on the final results. This uncertainty is determined by varying the binning of the mixed events in v_z and is correlated for different angles relative to the event plane and for different bins in $p_{T,\text{assoc}}$. Above $p_{T,\text{assoc}} > 3 \text{ GeV}/c$, this uncertainty is negligible because the background is small.

5.1.3 Flow modulation of combinatorial background

The combinatorial background ($N_{\text{assoc,jet}}^{\text{bkgd}}$) from Eq. 5.3 can be parametrized for trigger jets restricted to the orientation $\mathfrak{R} = \text{in/mid/out-of-plane}$ relative to the event plane as [13, 22]:

$$\left(\frac{1}{\pi} \frac{dN_{\text{assoc,jet}}^{\text{bkgd}}}{d\Delta\phi} \right)_{\mathfrak{R}} = B^{\mathfrak{R}} \left(1 + \sum_{n=2}^{\infty} 2v_{n,\text{assoc}} v_{n,\text{jet}}^{\mathfrak{R}} \cos(n\Delta\phi) \right), \quad (5.4)$$

where $v_{n,\text{assoc}}$ and $v_{n,\text{jet}}^{\mathfrak{R}}$ are the Fourier coefficients of the azimuthal angle distribution (flow coefficients) of background associated particles and trigger jets restricted to the orientation $\mathfrak{R}$ respectively. $B^{\mathfrak{R}}$ gives the background level amplitude for orientation $\mathfrak{R}$. The trigger jets being restricted to orientation $\mathfrak{R}$ modifies their nominal flow coefficients $v_{n,\text{jet}}$'s into the $v_{n,\text{jet}}^{\mathfrak{R}}$ from Eq. 5.4 which are related to each other as [13]:

$$v_{n,\text{jet}}^{\mathfrak{R}} = \begin{cases} \frac{v_{n,\text{jet}}}{\beta^{\mathfrak{R}}} + \cos(n\phi_s^{\mathfrak{R}}) \frac{\sin(nc)}{nc} \frac{R_n}{\beta^{\mathfrak{R}}} + \\ \frac{1}{\beta^{\mathfrak{R}}} \sum_{k=2,4,6,..} (v_{(k+n),\text{jet}} + v_{|k-n|,\text{jet}}) \cos(k\phi_s^{\mathfrak{R}}) \frac{\sin(kc)}{kc} R_n & \text{if } n \text{ is even,} \\ v_{n,\text{jet}} & \text{if } n \text{ is odd.} \end{cases} \quad (5.5)$$

Where, $\beta^{\mathfrak{R}}$ is the $\mathfrak{R}$ dependence of $B^{\mathfrak{R}}$ given by,

$$B^{\mathfrak{R}} \propto \beta^{\mathfrak{R}} = 1 + \sum_{k=2,4,6,..} 2v_{k,\text{jet}} \cos(k\phi_s^{\mathfrak{R}}) \frac{\sin(kc)}{kc} R_n \quad (5.6)$$

$\phi_s^{\mathfrak{R}}$ and c are the center and width of the $|\Psi_{2,EP} - \phi_{\text{jet}}|$ range for jets restricted to

orientation $\mathfrak{R}$. Odd $v_{n,\text{jet}}$'s mainly arise from initial state fluctuations and are therefore uncorrelated with the second-order event-plane and remain constant when the trigger jet is moved relative to the event plane, while even $v_{n,\text{jet}}$'s will change [22, 23]. The $\mathfrak{R}$ dependent background shape is dependent upon the event-plane resolution (R_n), which is fixed at the measured values. Extended details into the derivations of relevant equations can be found in Refs. [13, 24, 25].

Collective particle flow plays a major role in understanding the underlying-event background. This background consists of particles created from mechanisms unrelated to the hard process that led to a jet. Some of the jet signal is correlated with our event plane due to the path-length dependence of partonic energy loss, while soft hadrons are predominantly correlated with the event plane due to hydrodynamical flow that also contribute to the bulk-particle production.

To remove the combinatorial background comprised by contributions from the underlying event, the reaction plane fit (RPF) developed in Ref. [22] is applied in this work. The measured jet-hadron correlation signal is decomposed into a near-side and an away-side, with the former being narrow in $\Delta\phi$ and $\Delta\eta$ and the latter narrow in $\Delta\phi$, but broad in $\Delta\eta$. The narrowness of the near-side implies the signal is negligible at large $\Delta\eta$, where the background dominates. Applying RPF, we defined our ‘signal + background’ region for $|\Delta\eta| < 0.6$ and further $0.6 \leq |\Delta\eta| < 1.2$ as a background dominated region where the signal was assumed to be negligible. The correlation function is fit with the parametrization given in Eq. 5.4, restricted to $n = 4$ for the background dominated region at large $\Delta\eta$ and small $\Delta\phi$ ($|\Delta\phi| < \pi/2$) simultaneously for in-plane, mid-plane, and out-of-plane trigger jets.

RPF improves upon prior background subtraction techniques [6, 26] by avoiding problems due to contamination from jets on both the near- and away-side by using the near-side at large $\Delta\eta$ and also using the dependence of the flow-modulated background on the angle of the trigger jet relative to the event plane to constrain the background shape and level.

For $p_{T,\text{assoc}} > 2 \text{ GeV}/c$, the combinatorial background is small, and few high momentum

tracks are found at large distances in pseudorapidity from the near-side jet. So the model fit is restricted to $n = 3$ as higher order terms are no longer contributing. This occurs for $p_{T,assoc} > 4$ GeV/ c with $15 < p_{T,jet} < 20$ GeV/ c jets and $p_{T,assoc} > 3$ GeV/ c with $20 < p_{T,jet} < 40$ GeV/ c . Therefore, the RPF fits consist of six parameters ($B^{\mathfrak{R}}$, $v_{2,jet}^{\mathfrak{R}}$, $v_{2,assoc}$, $(v_{3,jet} \times v_{3,assoc})$, $v_{4,jet}^{\mathfrak{R}}$, and $v_{4,assoc}$) for low $p_{T,assoc}$, and four ($B^{\mathfrak{R}}$, $v_{2,jet}^{\mathfrak{R}}$, $v_{2,assoc}$ and $(v_{3,jet} \times v_{3,assoc})$) for high $p_{T,assoc}$.

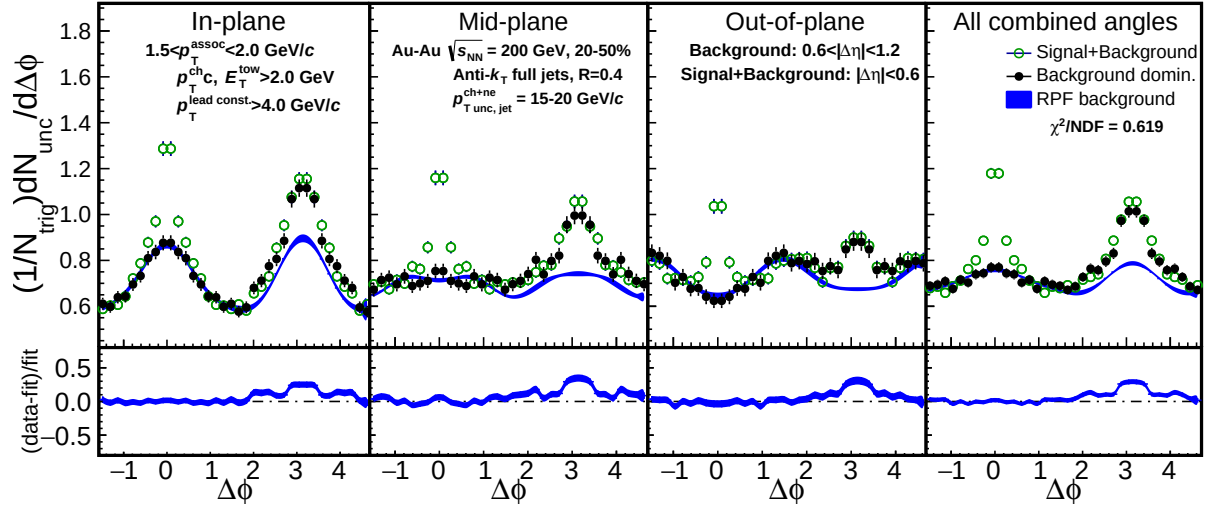


Figure 5.2: (Top) Signal+background region, background-dominated region, and RPF fit to the background for the event-plane dependent jet-hadron correlations of $15 < p_{T,jet}^{\text{jet}} < 20$ GeV/ c jets correlated with $1.5 < p_{T,assoc} < 2.0$ GeV/ c charged hadrons from the 20-50% most central events. (Bottom) Quality of the RPF fit to the background-dominated region expressed as (data - fit) / fit.

Comparison of in-, mid-, and out-of-plane jet-hadron correlations is performed after background subtraction to explore the effects related to event-plane orientation. An example of event-plane dependent correlation function after the RPF background subtraction for jets with $15 < p_{T,jet} < 20$ GeV/ c is shown in Fig. 5.3 for in-plane (a), mid-plane (b), out-of-plane (c) and jets from all combined angles (d) for associated particles with momenta $1.5 < p_{T,assoc} < 2.0$ GeV/ c . The uncertainties from the RPF background subtraction are propagated using the covariance matrix from the fit and are non-trivially correlated point-to-point and between different bins relative to the event plane. These are shown in gray.

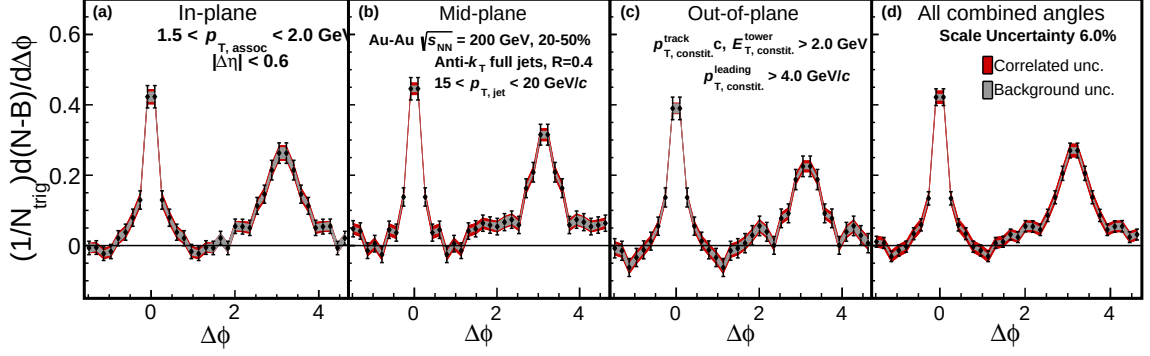


Figure 5.3: Example of a background subtracted correlation function for $15 < p_{T,jet} < 20$ GeV/ c jets correlated with $1.5 < p_{T,assoc} < 2.0$ GeV/ c charged hadrons from the 20-50% most central collisions. Correlated scaled uncertainty from the application of mixed events is displayed by the red band, while the uncertainty associated with the RPF background fit is displayed as the gray band.

The uncertainty from the acceptance correction, described in Sect. 5.1.2 is displayed by the red uncertainty band. The uncertainties on the event-plane resolution are negligible relative to that of the background subtraction and statistical uncertainties of the final results. Additional uncertainties uncorrelated with each other, but correlated for all points in $\Delta\phi$ are given in Tab. 5.1. They are combined in quadrature and listed as the scale uncertainty on the results.

5.1.4 Systematic Uncertainties

The PYTHIA6 Perugia 2012 tune is used to create particle level dijet events embedded in MB Au+Au 200 GeV events at the detector level. This allows for further comparisons between the jets from the input PYTHIA6 tracks (generator-level jets or GEN-jets) and the jets from the embedded tracks reconstructed by GEANT (reconstructed jets or RECO-jets). RECO level events are analyzed with the same selections and parameters as used by the data analysis. All particles of GEN-level events are required to be in their final state (particles with no further daughters). GEN-level jets have $p_{T,jet}^{GEN} > 10$ GeV/ c and the RECO-level jets have $p_{T,jet}^{RECO} > 5$ GeV/ c . Further, we require only the RECO-level jets contain a neutral component with $E_T \geq 5.4$ GeV to match the trigger condition applied in data. Our goal is

to study the effects of the detector response on jet reconstruction and our analysis. In order to compare the same jet pre- and post-reconstruction, we apply a ‘nearness’ criteria in the $\eta - \phi$ that matches a GEN-jet to one RECO-jet satisfying:

(a) $r_{\text{GEN,RECO}} = \sqrt{(\eta_{\text{jet}}^{\text{GEN}} - \eta_{\text{jet}}^{\text{RECO}})^2 + (\phi_{\text{jet}}^{\text{GEN}} - \phi_{\text{jet}}^{\text{RECO}})^2} \leq 0.4$, $r_{\text{GEN,RECO}}$ being the separation between the gen-jet and the reco-jet in $\eta - \phi$ space, (b) the RECO-jet is the closest to the gen-jet among all reco-jets (minimize $r_{\text{GEN,RECO}}$).

We compare the resulting GEN- and RECO-level jet spectra to calculate the momentum resolution which is given by:

$$p_{\text{T,jet}} \text{ resolution}(\%) = \frac{p_{\text{T,jet}}^{\text{GEN}} - p_{\text{T,jet}}^{\text{RECO}}}{p_{\text{T,jet}}^{\text{GEN}}} \times 100\%. \quad (5.7)$$

Fig. 5.4 shows the $p_{\text{T,jet}}$ resolution for $15 < p_{\text{T,jet}}^{\text{GEN}} < 20$ GeV/ c (left) and $20 < p_{\text{T,jet}}^{\text{GEN}} < 40$ GeV/ c (right) $R = 0.4$ full jets. The distributions have been normalized into probability functions and are shown for the 20-50% most central events for all angles of the jet relative to the event plane. Fig. 5.4 further shows an average energy loss of around 15% going from GEN to RECO. This net energy shift is thought to be due to counteracting effects of the tracking inefficiency at the RECO level and there being more tracks in the RECO level from the min-bias pedestal. The $p_{\text{T,jet}}$ resolution for $15 < p_{\text{T,jet}}^{\text{RECO}} < 40$ GeV/ c reco-jets is roughly 10-20%. The event-plane dependence of the $p_{\text{T,jet}}$ resolution was also studied and found to be within 1-2% of each other between different orientations of jets w.r.t event plane. There can be slight differences in the jets reconstructed at lower momenta with $15 < p_{\text{T,jet}} < 20$ GeV/ c for jets at different angles relative to the event plane, due to a low momentum embedded jet overlapping with another jet in the Au+Au data and from statistical fluctuations. As there are more jets from in-plane than out-of-plane orientations in the data, this leads to an apparent difference in the reconstructed jet spectra. Otherwise there are no significant differences between jets at different angles relative to the event plane. A map between $p_{\text{T,jet}}^{\text{GEN}}$ and $p_{\text{T,jet}}^{\text{RECO}}$, called the response matrix, is added to Fig. 5.4 as a smaller inset on the top

right.

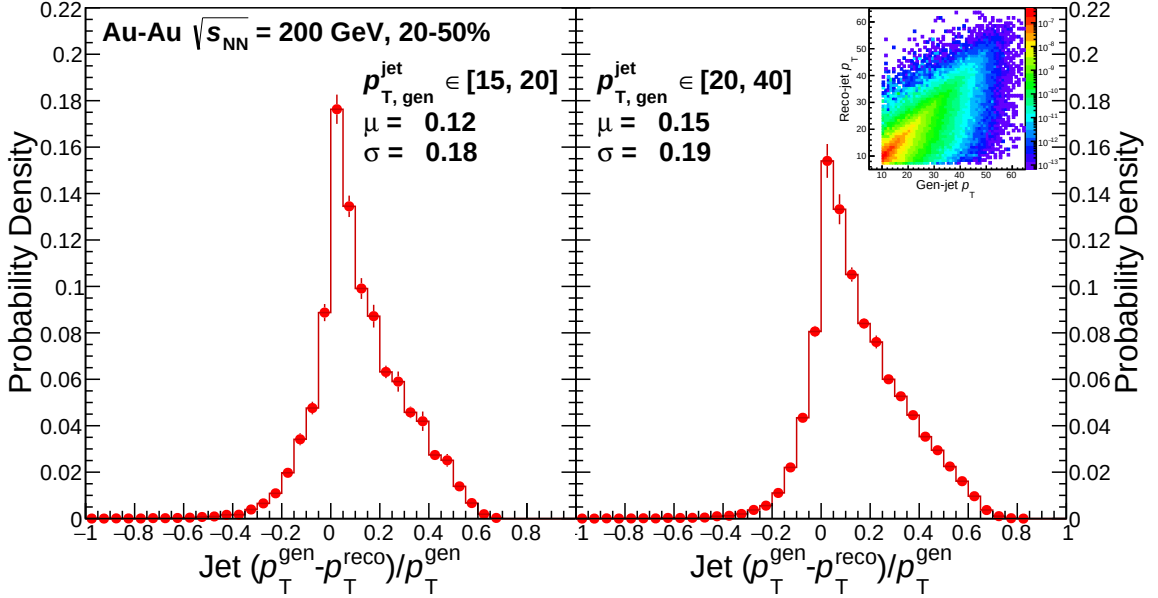


Figure 5.4: $p_{T,jet}$ resolution for $15 < p_{T,jet}^{GEN} < 20$ GeV/ c (left) and $20 < p_{T,jet}^{GEN} < 40$ GeV/ c (right) $R = 0.4$ full jets with the corresponding response matrix as an inset on the top right. Jets are measured from all angles relative to the event plane in the 20-50% most central events. This comparison is for matched GEN-level to RECO-level jets.

We summarize the systematic uncertainties in Tabs. 5.1, 5.2, and 5.3. Table 5.1 lists the sources of systematic uncertainties which are independent of the angle relative to the event plane. These sources include the single-track reconstruction efficiency (Sect. ??), uncertainties in the mixed events due to their normalization and shape in $\Delta\phi$ (Sect. 5.1.2), and uncertainties in the event-plane resolution (Sect. 3.2.2). The uncertainties are added in quadrature and lead to a 6% uncertainty in the scale of the correlation functions and yields with the single-track reconstruction efficiency being the dominate source. This uncertainty is uncorrelated for different associated-particle momenta.

Additional uncertainties, highly dependent on the angle of the jet relative to the event plane and the associated particles' momentum, are summarized in Tabs. 5.2 and 5.3. These include the impact of the scale uncertainty from the mixed events (Sect. 5.1.2) and the RPF background fit (Sect. 5.1.3) on the associated yield and jet-peak width results. These uncertainties are compared for two associated particle momentum bins (1.0-1.5 and 3.0-4.0

Table 5.1: Summary of systematic uncertainties which are independent of the angle relative to the event plane and the momentum for both $15 < p_{T,\text{jet}} < 20$ GeV/ c and $20 < p_{T,\text{jet}} < 40$ GeV/ c in 20-50% central Au+Au collisions.

Source	Uncertainty %
Single particle reconstruction efficiency	5
Mixed event (shape $\Delta\phi$)	negligible
Mixed event normalization	< 1.25
Event-plane resolution	< 1

GeV/ c) to highlight how much of an impact the background has at low momenta. The uncertainty arising from the RPF background subtraction is non-trivially correlated point-to-point in $\Delta\phi$ and for different orientations of the jet relative to the event plane, but is uncorrelated between different $p_{T,\text{assoc}}$ bins. The acceptance-correction uncertainty on the shape in $\Delta\eta$ is also correlated for different orientations of the jet relative to the event plane and uncorrelated across $p_{T,\text{assoc}}$ bins. The acceptance-shape uncertainty is the dominant source at low-momenta, while being more comparable to the background uncertainty at larger momenta.

To account for jet-energy shift (JES) due to underlying-event background contribution to the reconstructed jet energy and to provide an adequate comparison for the jet samples from different orientations relative to event plane, a random-cone study was performed on the minimum bias data of matching centrality selection. The $\langle\rho\rangle$ is found to be 0.388 GeV/ c (in-plane), 0.344 GeV/ c (mid-plane), and 0.308 GeV/ c (out-of-plane), with a corresponding RMS of 0.4 GeV/ c . Shifting the jet-momenta selection by the corresponding thresholds was utilized in the correlation analysis to remove the shift-related differences between the jet collections.

5.2 Results

Charged particle yields associated with jets are found by integrating the associated yield:

$$\text{Yield} = \frac{1}{N_{\text{trig}}} \int_c^d \int_a^b \frac{d^2 N_{\text{assoc,jet}}}{d\Delta\phi d\Delta\eta} d\Delta\phi d\Delta\eta. \quad (5.8)$$

Table 5.2: Summary of systematic uncertainties on the associated yields and widths calculated from the correlation functions due to the shape uncertainty coming from the shape of the acceptance correction in $\Delta\eta$, the correlated background-fit uncertainty, and the uncertainty associated with the jet energy shift (JES) correction, each varying with event-plane orientation bins. They are displayed for $15 < p_{T,\text{jet}} < 20$ GeV/ c in 20-50% central Au+Au collisions for $1.0 < p_{T,\text{assoc}} < 1.5$ GeV/ c and $3.0 < p_{T,\text{assoc}} < 4.0$ GeV/ c bins. The values are expressed as a percent of the nominal value.

Source	Result	Orientation	Uncertainty %			
			Near-side: $p_{T,\text{assoc}}$ (GeV/ c)		Away-side: $p_{T,\text{assoc}}$ (GeV/ c)	
			1.0-1.5	3.0-4.0	1.0-1.5	3.0-4.0
Acceptance shape	Yield	in-plane	14	1.2	8.1	3.0
		mid-plane	11	1.2	7.6	3.3
		out-of-plane	11	1.1	7.3	3.0
	Width	in-plane	5.2	0.6	4.3	2.4
		mid-plane	5.4	0.5	3.9	2.2
		out-of-plane	4.3	0.4	4.9	2.1
Background fit	Yield	in-plane	11	0.8	6.1	2.1
		mid-plane	7.7	0.9	5.2	2.5
		out-of-plane	7.4	0.7	4.7	2.1
	Width	in-plane	10	0.1	8.2	0.4
		mid-plane	10	0.1	7.5	0.4
		out-of-plane	8.2	0.1	9.3	0.4
JES correction	Yield	in-plane	1.8	3.7	4.9	4.0
		mid-plane	2.8	2.5	1.4	4.2
		out-of-plane	3.7	3.3	4.8	4.2
	Width	in-plane	0.6	0.4	4.0	< 0.1
		mid-plane	0.2	< 0.1	7.9	1.5
		out-of-plane	1.6	0.7	1.9	1.7

Table 5.3: Summary of systematic uncertainties on the associated yields and widths calculated from the correlation functions due to the shape uncertainty coming from the shape of the acceptance correction in $\Delta\eta$, the correlated background-fit uncertainty, and the uncertainty associated with the JES correction, each varying with event-plane orientation bins. They are displayed for $20 < p_{T,\text{jet}} < 40$ GeV/ c in 20-50% central Au+Au collisions for $1.0 < p_{T,\text{assoc}} < 1.5$ GeV/ c and $3.0 < p_{T,\text{assoc}} < 4.0$ GeV/ c bins. The values are expressed as a percent of the nominal value.

Source	Result	Orientation	Uncertainty %			
			Near-side: $p_{T,\text{assoc}}$ (GeV/ c)		Away-side: $p_{T,\text{assoc}}$ (GeV/ c)	
			1.0-1.5	3.0-4.0	1.0-1.5	3.0-4.0
Acceptance shape	Yield	in-plane	35	1.2	22	2.5
		mid-plane	39	1.2	22	2.2
		out-of-plane	59	0.9	37	1.6
	Width	in-plane	30	0.7	8.4	1.9
		mid-plane	22	0.5	15	1.7
		out-of-plane	19	0.4	28	1.2
Background fit	Yield	in-plane	13	1.0	8.2	2.1
		mid-plane	14	0.7	8.0	1.2
		out-of-plane	22	0.9	14	1.5
	Width	in-plane	13	0.1	3.7	0.2
		mid-plane	10	0.1	6.8	0.2
		out-of-plane	8.6	< 0.1	13	0.1
JES correction	Yield	in-plane	0.2	0.4	< 0.1	2.6
		mid-plane	1.1	0.3	2.1	2.9
		out-of-plane	3.1	0.6	1.4	3.5
	Width	in-plane	1.2	0.5	3.3	0.3
		mid-plane	1.2	0.7	1.3	1.5
		out-of-plane	2.2	0.1	0.2	1.9

The integration limits in $\Delta\phi$ for the near-side are $a = -\pi/3$ and $b = \pi/3$, while for the away-side, we have $a = 2\pi/3$ and $b = 4\pi/3$. The integration limits in $\Delta\eta$ are the same for both the near-side and away-side, $c = -0.6$ and $d = 0.6$.

Shown in Fig. 5.5 is the near-side (left) and away-side (right) associated yields vs $p_{T,\text{assoc}}$ for $15 < p_{T,\text{jet}} < 20$ GeV/ c (top) and $20 < p_{T,\text{jet}} < 40$ GeV/ c (bottom) full jets in 20-50% centrality collisions. The yields are compared for each orientation of the trigger jet reconstructed relative to the event plane (in/mid/out) for $p_{T,\text{assoc}} \in [1.0, 1.5], [1.5, 2.0], [2.0, 3.0], [3.0, 4.0], [4.0, 6.0], [6.0, 10.0]$ GeV/ c .

The main feature is the steeply falling associated yield with increasing $p_{T,\text{assoc}}$, which occurs on both the near- and away-side. Note that associated yields for $p_{T,\text{assoc}} \geq 2.0$ GeV/ c include jet constituents, which leads to the discontinuity at $p_{T,\text{assoc}} = 2$ GeV/ c . Additionally, the use of jets with ‘hard-cores’ and containment of a tower associated to the firing event trigger can lead to a surface bias of the near-side jet. This, however, maximizes the average path length the away-side recoil jets travels, increasing the likelihood of an interaction with the medium. We would expect an in-plane jet and an out-of-plane jet with the same $p_{T,\text{jet}}$ to have different distributions of hard and soft constituents. An in-plane jet with less path travelled in the medium on average would be expected to show higher yields of constituents with higher $p_{T,\text{assoc}}$ than the more quenched out-of-plane jet, which would be expected to show higher yields of constituents with lower $p_{T,\text{assoc}}$. While uncertainties are smaller for jets with $15 < p_{T,\text{jet}} < 20$ GeV/ c , both samples still lack any clear dependence on the event-plane angle. This is an indication that modifications dependent on the average path length are smaller than the experimental uncertainties. On the far right of the near- and away-side panels is the inclusive $p_{T,\text{assoc}}$ bin with $1.0 < p_{T,\text{assoc}} < 10$ GeV/ c . The associated yields of each event-plane orientation in the inclusive $p_{T,\text{assoc}}$ selection are consistent with each other for the sample where $15 < p_{T,\text{jet}} < 20$ GeV/ c . However, in the jet sample with $20 < p_{T,\text{jet}} < 40$ GeV/ c , there are indications suggesting potential modifications. This potential modification is apparent on both the near- and away-side with the largest

contributions coming from the lowest $p_{T, \text{assoc}}$ bins.

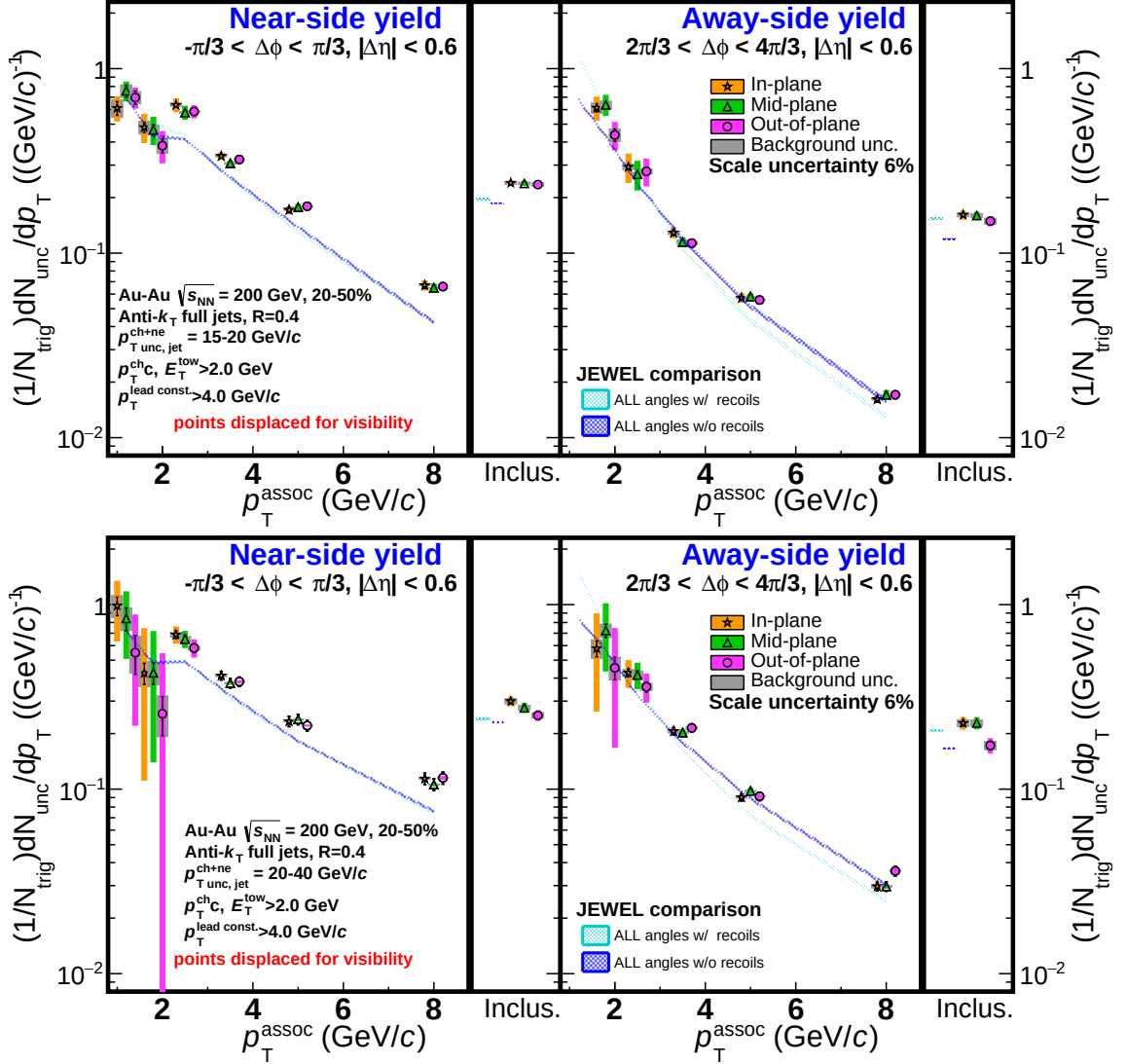


Figure 5.5: Near-side (left) and away-side (right) uncorrected (unc.) associated yield vs $p_{T, \text{assoc}}$ for 15-20 (top) and 20-40 GeV/c (bottom) full jets of 20-50% centrality in Au+Au collisions. The grey bands describe the systematic uncertainties of the background fits and are non-trivially correlated point-to-point. The colored bands are scale uncertainties from the mixed event acceptance shape and JES correction. There is an additional 6% global scale uncertainty. Included on the far right of the near- and away-side panels is the inclusive transverse momentum bin from 1.0-10.0 GeV/c . Points are displaced for visibility.

The widths are calculated by fitting a Gaussian, $Ae^{(\Delta\phi - \Delta\phi_0)^2/2\sigma^2}$, to the jet peak centered at $\Delta\phi_0 = 0$ for the near-side and $\Delta\phi_0 = \pi$ for the away-side. The gaussians are fitted separately, with a range of $|\Delta\phi| < \pi/3$ on the near-side and $|\Delta\phi - \pi| < \pi/3$ on the

away-side. The Gaussian fit is repeated with different values of the background parameters and the covariance matrix is used to propagate the uncertainties. The scale uncertainties on the widths are given by $\sigma_w^{sc} = \frac{B}{\sigma_B} \times |\alpha - 1|$, where α is the p_T -dependent scale factor associated with the acceptance shape when propagating the background determined in the $0.6 \leq |\Delta\eta| < 1.2$ to the region $|\Delta\eta| < 0.6$.

From Fig. 5.6, it is clear a broadening of the jet peaks is occurring for decreasing $p_{T, \text{assoc}}$. This is expected from both collisional energy loss and gluon bremsstrahlung. With out-of-plane jets expected to traverse a longer average path length than in-plane jets, this would lead to additional interactions with the medium and more subsequent re-scatterings resulting in a larger width for jets out-of-plane relative to in-plane. Within the uncertainties there is no clear ordering. This indicates the effect of path-length dependent energy loss is not large enough to be seen by the current precision of the data. On the far right of the near- and away-side panels is the inclusive $p_{T, \text{assoc}}$ bin with $1.0 < p_{T, \text{assoc}} < 10 \text{ GeV}/c$. The widths of each event-plane orientation in the inclusive selection for the the sample with $15 < p_{T, \text{jet}} < 20 \text{ GeV}/c$ are consistent. Conversely, the $15 < p_{T, \text{jet}} < 20 \text{ GeV}/c$ sample reveals indications of potential modifications. This potential modification is apparent on both the near- and away-side and primarily comes from the lowest transverse momentum bins where sample size is the largest.

The measurements presented in Figs. 5.5 and 5.6 are compared to calculations from “Jet Evolution With Energy Loss” known as JEWEL [27], a jet energy loss model based on radiative and collisional energy loss in connection with partons sampled from a longitudinally expanding medium [28]. Model calculations are provided for two regimes, for calculations that a) include recoiled partons, and b) do not include recoiled partons. When no recoils are kept, the lost jet momentum is removed from the entire system. This is useful for modeling energy loss in the hard part of the jet. When recoils are kept, the jets momentum is fully conserved, but this adds both energy and additional background particles to the di-jet. In an experimental analysis, we likely would measure some, but not all of the recoil particles

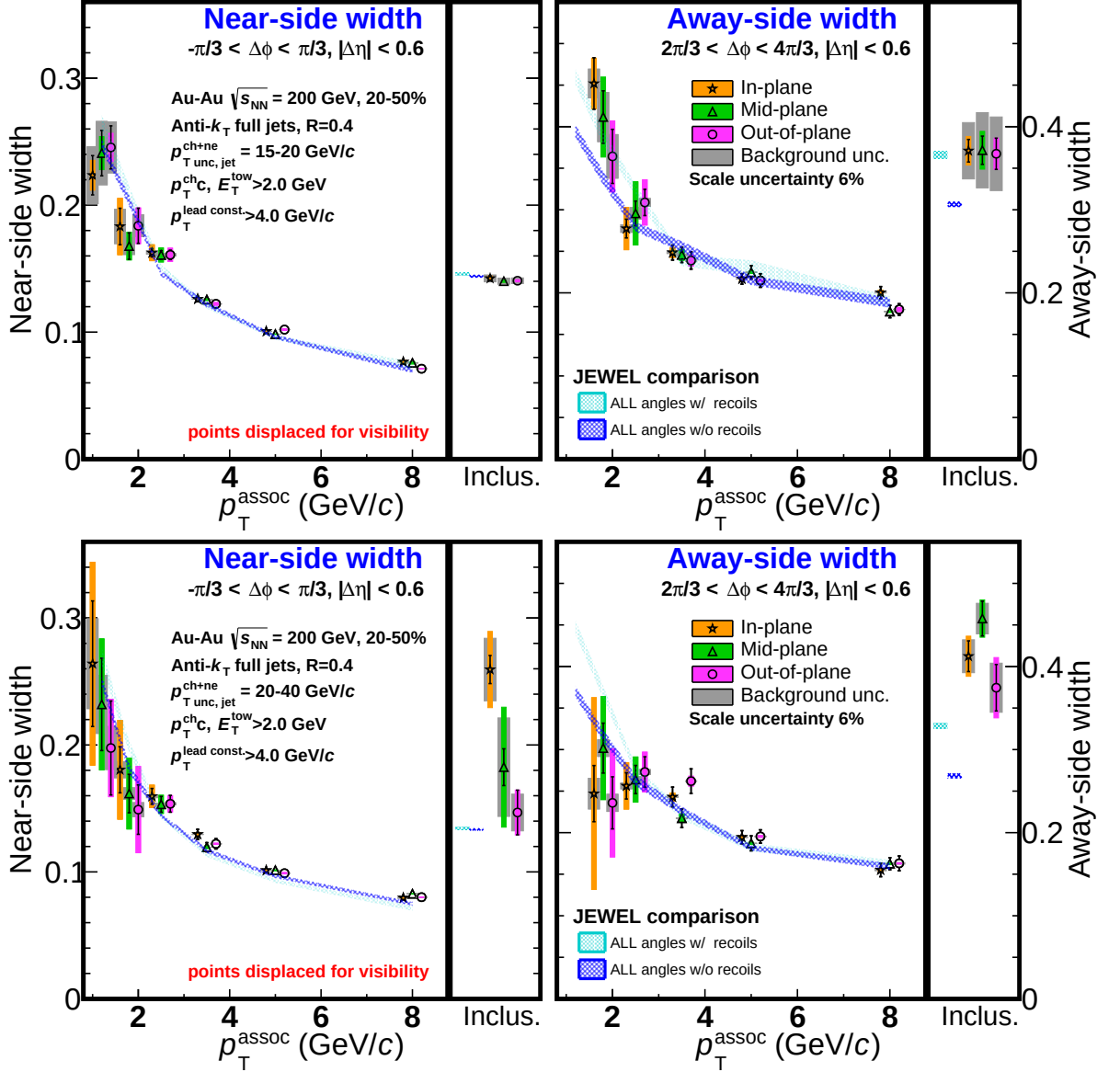


Figure 5.6: Near-side (left) and away-side (right) jet width vs $p_{T,assoc}$ for 15-20 (top) and 20-40 GeV/c (bottom) full jets of 20-50% centrality in Au+Au collisions. The widths are extracted from the gaussian fit to the jet peak. The grey bands describe the systematic uncertainties of the background fits which are non-trivially correlated point-to-point. The colored bands are scale uncertainties from the mixed-event acceptance shape and JES correction. There is an additional 6% global scale uncertainty. Included on the far right of the near- and away-side panels is an inclusive transverse momentum bin from 1.0-10.0 GeV/c. Points are displaced for visibility.

as they are often indistinguishable from background.

Due to the dominant impact of jet-by-jet fluctuations on partonic energy loss over path-length dependence [29, 30], JEWEL only predicts a very slight event-plane dependence which is well below the systematic uncertainty in the measurement. Variations among event-plane orientations were not seen at the 10% level. This is therefore consistent with path-length dependence having an insignificant impact compared to jet-by-jet fluctuations in energy loss. Fluctuations in the density of the medium may also suppress observable path-length dependence and are not included in the JEWEL model. However, higher precision JEWEL calculations may be needed to discern any potential event-plane dependent effects. We thus show the JEWEL comparisons corresponding to the sample integrated over all angles relative to the event plane and compare to the results from data. Comparisons show that the away-side is well described in terms of both the associated yields and widths by including recoils at low $p_{T,\text{assoc}}$, while at high $p_{T,\text{assoc}}$, the yields are better described by not including recoils and the widths have similar results to within uncertainties for both cases. When looking at the near-side, the widths are quite similar at high $p_{T,\text{assoc}}$ with slightly larger values when including recoils at low $p_{T,\text{assoc}}$. For the associated yields at high $p_{T,\text{assoc}}$, both JEWEL cases are similar to each other, but underestimate the data and including recoils has larger yields that better match the data at low $p_{T,\text{assoc}}$.

Alternatively, the initial geometry within specific centrality bins may be fluctuating at a larger magnitude than any possible event-plane dependence [31, 32]. The possibility of this occurrence can be studied by looking at the initial configuration and selecting low and high ellipticity events by using the q_n flow vector found within a selected centrality range.

To study the impact of surface bias and event-plane resolution, a check was performed to investigate the systematic change in the ratio of yields (out/in, mid/in) with the angle of the event plane by fitting a constant to Fig. ?? for jets with $15 < p_{T,\text{jet}} < 20$ GeV/ c (top) and $20 < p_{T,\text{jet}} < 40$ GeV/ c (bottom). The systematic uncertainties are treated as uncorrelated point-to-point and added to the statistical uncertainties in quadrature. The results are shown in Tabs. 5.4 and 5.5 and are consistent with one. Care should be taken when

interpreting the results, as various effects can be in play and medium modifications could give way to a $p_{T,\text{assoc}}$ dependence. This $p_{T,\text{assoc}}$ dependence can be seen as an enhancement of associated yields for $p_{T,\text{assoc}} \geq 2$ GeV/ c on the near-side from the geometric kinematic biases due to the “hard-core” requirement of jet reconstruction.

Table 5.4: Results of fits to Fig.?? (top panel: 15-20 GeV/ c jets) to a constant c , the χ^2 over the number of degrees of freedom (NDF), the number of standard deviations σ of c from one, and the range of c within a 90% confidence limit (CL).

parameter	Near-side		Away-side	
	$Y_{\text{out}}/Y_{\text{in}}$	$Y_{\text{mid}}/Y_{\text{in}}$	$Y_{\text{out}}/Y_{\text{in}}$	$Y_{\text{mid}}/Y_{\text{in}}$
c	0.93 ± 0.042	0.949 ± 0.038	0.89 ± 0.065	0.997 ± 0.066
χ^2/NDF	0.68	0.67	0.71	0.18
σ	-1.6	-1.3	-1.7	-0.1
90% CL	$0.86 - 1.00$	$0.89 - 1.01$	$0.78 - 1.00$	$0.94 - 1.05$

Table 5.5: Results of fits to Fig.?? (bottom panel: 20-40 GeV/ c jets) to a constant c , the χ^2 over the number of degrees of freedom (NDF), the number of standard deviations σ of c from one, and the range of c within a 90% confidence limit (CL).

parameter	Near-side		Away-side	
	$Y_{\text{out}}/Y_{\text{in}}$	$Y_{\text{mid}}/Y_{\text{in}}$	$Y_{\text{out}}/Y_{\text{in}}$	$Y_{\text{mid}}/Y_{\text{in}}$
c	0.874 ± 0.043	0.937 ± 0.043	0.752 ± 0.064	1.02 ± 0.075
χ^2/NDF	1.4	0.19	2.1	0.49
σ	-2.9	-1.5	-3.9	0.2
90% CL	$0.77 - 0.98$	$0.90 - 0.97$	$0.56 - 0.94$	$0.91 - 1.12$

5.3 Conclusions

The measurement of jet-hadron correlations relative to the event plane is reported for the 20-50% most central events in Au+Au collisions at $\sqrt{s_{\text{NN}}} = 200$ GeV in STAR. Partonic interactions are directly related to the distance traversed in the medium, so it is expected that medium-induced jet modifications should depend on the path length. The angle of the jet, measured with respect to the event plane, is correlated on average with the jet’s path length through the medium. In this analysis, the average path length of away-side jets is potentially increased due to the surface bias of the near-side trigger jet.

This work utilizes the RPF background-subtraction method to remove the event-plane dependent background while reducing uncertainties and assumptions associated with previous background-subtraction techniques. Associated yields, their ratios, and jet-peak widths are extracted for each event-plane orientation and compared with different average path lengths and JEWEL model calculations. JEWEL performs better in describing the associated yields and widths at higher $p_{T,\text{assoc}}$ when recoil partons are not included. Conversely, including recoil partons leads to JEWEL providing a better description of the lower $p_{T,\text{assoc}}$ region. This study highlights the importance of conducting further tuning of Monte Carlo simulations to accurately describe the results in this analysis for jets that are biased towards hard-fragmented jets due to the HT-trigger and hard-core constituent requirements.

Chapter 6

Generalized Angularities

Generalized jet angularities are a family of jet observables defined as,

$$\lambda_\beta^\kappa = \sum_{\text{const} \in \text{jet}} \left(\frac{p_{\text{T,const}}}{p_{\text{T,jet}}} \right)^\kappa \left(\frac{\Delta R_{\text{const,jet}}}{R} \right)^\beta \quad (6.1)$$

where $p_{\text{T,jet/const}}$ is the jet/constituent's momentum, and $\Delta R_{\text{const,jet}}$ is the (η, ϕ) distance of a constituent from the jet-axis given by,

$$\Delta R_{\text{const,jet}} = \sqrt{(\eta_{\text{jet}} - \eta_{\text{const}})^2 + (\phi_{\text{jet}} - \phi_{\text{const}})^2} \quad (6.2)$$

Parameters κ and β tune experimental sensitivity to hard and wide-angle radiation, respectively. λ_β^1 's are infra-red and collinear (IRC) safe angularities [33], which probe the average angular spread of energy around the jet-axis. They are radial moments of the jet's transverse momentum profile, also known as differential jet-shape ($\pi_T(\mathbf{r})$), given by,

$$\pi_T(\mathbf{r}) = \lim_{\delta r \rightarrow 0} \left\langle \frac{1}{\delta r} \frac{\sum_{|\mathbf{r}_{\text{const}} - \mathbf{r}| < \delta r/2} p_{\text{T,const}}}{p_{\text{T,jet}}} \right\rangle_{\text{jets}}, \quad (6.3)$$

where $\mathbf{r}_{\text{const}} = (\eta_{\text{const}} - \eta_{\text{jet}})\hat{\eta} + (\phi_{\text{const}} - \phi_{\text{jet}})\hat{\phi}$, and it follows that,

$$\lambda_\beta^1 = \int_{\text{jet}} r^\beta \pi_T(\mathbf{r}) d\mathbf{r}. \quad (6.4)$$

However, as an experimental observable, it is useful to restrict the summation in Eq. 6.1 to only charged constituents / tracks and mitigate contamination from towers like pileup.

This comes at the cost of IRC safety of λ_β^1 as theoretical predictions will involve non-perturbative fragmentation functions.

Using Recursive SoftDrop, we can decompose angularities into pQCD and npQCD components. pQCD component is the angularity calculated from the jet after grooming has been performed and the npQCD component is calculated as the difference between the inclusive/ungroomed and groomed jet angularities.

$$\begin{aligned}
\lambda_\beta^\kappa &= \lambda_{\beta,\text{pQCD}}^\kappa + \lambda_{\beta,\text{npQCD}}^\kappa \\
\lambda_{\beta,\text{pQCD}}^\kappa &\equiv \lambda_{\beta,g}^\kappa \\
\lambda_{\beta,\text{npQCD}}^\kappa &\equiv \Delta\lambda_\beta^\kappa = \lambda_\beta^\kappa - \lambda_{\beta,g}^\kappa
\end{aligned} \tag{6.5}$$

Using the momentum balance (z_g) and angularity of the first collinear-splitting (R_g), we get direct connection between measurement of jet shapes and calculations from field theory. Larger R_g implies that the most prominent split in the jet occurred earlier in its evolution, opening up more phase space for pQCD splittings (which survive SoftDrop grooming) because of color coherence discussed earlier. Fig. 6 shows percentage npQCD contribution to jet girth $\left(\frac{\Delta\lambda_1^1}{\lambda_1^1} \times 100\right)$ decreasing with increasing R_g , going from highly collimated, pencil-like jets to jets with proper two pronged structure.

6.1 Angularities of $p + p$ jets

6.1.1 Dataset and Selections

The analysis uses data from $p + p$ collisions at $\sqrt{s} = 200$ GeV collected in 2012 using the STAR detector system. Charged-particle tracks and neutral energy depositions (towers) are measured using STAR's Time Projection Chamber (TPC) [34] and Barrel Electromagnetic Calorimeter (BEMC) [4] detectors respectively.

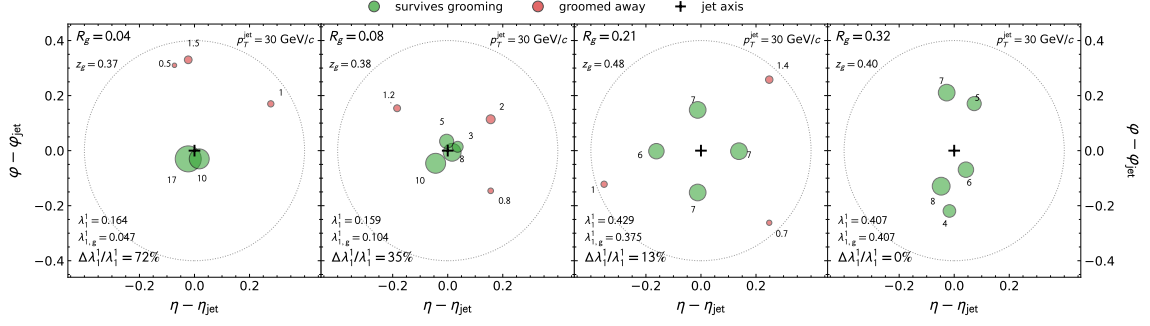


Figure 6.1: Ways of arranging constituents (each represented as a circle whose radius represent $\frac{p_{T,const}}{p_{T,jet}}$) in the (η, ϕ) space within a jet with $p_{T,jet} = 30 \text{ GeV}/c$ such that R_g of the jet increases going from left to right. Green (Red) colored constituents survive (are removed by) the SoftDrop grooming ($z_{cut} = 0.2$, $\beta = 0$).

To enhance jet signal, only High-Tower (HT) triggered events, with at least one tower with $E_{T,tower} \geq 4 \text{ GeV}$ are considered. Events containing a track with $p_{T,track} \geq 30 \text{ GeV}/c$ are rejected to avoid contamination from cosmic rays. Tracks(Towers) with $p_{T,track} \geq 0.2 \text{ GeV}/c$ ($E_{T,tower} \geq 0.2 \text{ GeV}$) are considered for jet clustering. Clustered jets completely falling within acceptance ($|\eta_{jet}| \leq 0.6$), with $p_{T,jet} \geq 10 \text{ GeV}/c$ and a plurality of charged tracks ($N_{ch,jet} \geq 2$) are used for further study.

The tracks and towers are clustered into jets using the anti- k_T algorithm with a jet resolution parameter $R = 0.4$, implemented using the FastJet library [35]. After obtaining the jets, the softer constituents are groomed using Recursive SoftDrop with $\beta = 0$ and $z_{cut} = 0.2$. The groomed-jet's constituents lose non-kinematic information like charge, which is regained by matching them to the original jet's constituents.

6.1.2 Detector effects and Unfolding

The embedding simulation starts from PYTHIA6 (STAR Tune) at the particle level. The particles are then reconstructed as detector hits using the STAR detector system implemented in GEANT3 and embedded into zero-bias events from 200 GeV $p + p$ collisions (2012). The jets from particle and detector levels are matched using a $p_{T,jet}$ ordered nearness criteria. The matched jets can be used to show detector responses like shown in Fig.

6.1.2

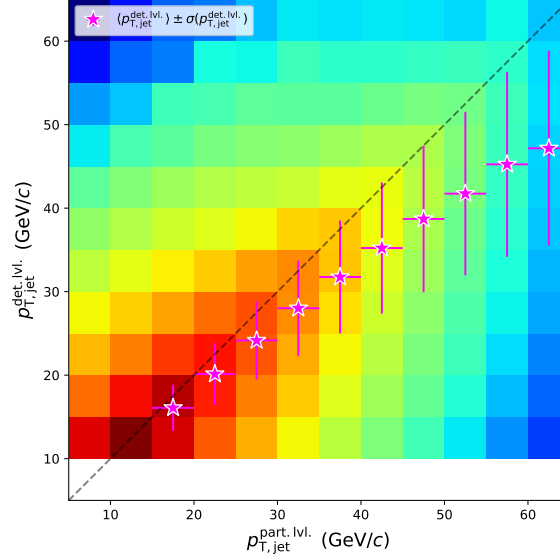


Figure 6.2: Two-dimensional histogram showing detector response for $p_{T,jet}$, showing probability of a $p_{T,jet}^{GEN}$ particle level jet being reconstructed as $p_{T,jet}^{RECO}$. Purple stars shows expected $p_{T,jet}^{RECO}$ given a particular bin of the $p_{T,jet}^{GEN}$.

$p_{T,jet}$ -resolution shown in Fig. 6.1.2.

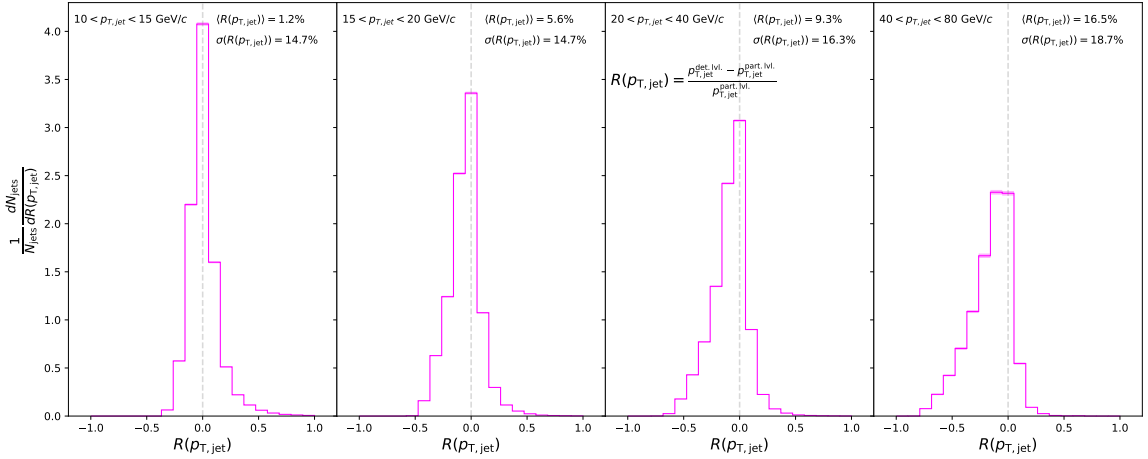


Figure 6.3: $p_{T,jet}$ -resolution in different $p_{T,jet}$ bins.

Unfolding is done using the Multifold method. Every jet is represented as a vector of 23 features. The four classifiers of Multifold are parametrized as DNNs with 3 hidden layers with 256 nodes each. The model weights are trained using the AdamW optimizer with a learning rate of 10^{-3} and weight decay of 10^{-2} . Convergence is achieved in two unfolding

iterations. Statistical uncertainties are calculated using bootstrap ensemble with 50 model replicas each sub-sampling 500k each of the data-like and sim-like jets, so each model replica sees 1M jets every epoch. All the required unfolded distributions and profiles are calculated by using the per-jet unfolding weights.

Closure test of the unfolding shown in Fig. 6.1.2 is done by doing the same unfolding procedure replacing measured data with pseudo-data for which the particle level is known (pseudo-truth). The pseudo-unfolded distributions are then compared to the pseudo-truth via ratios

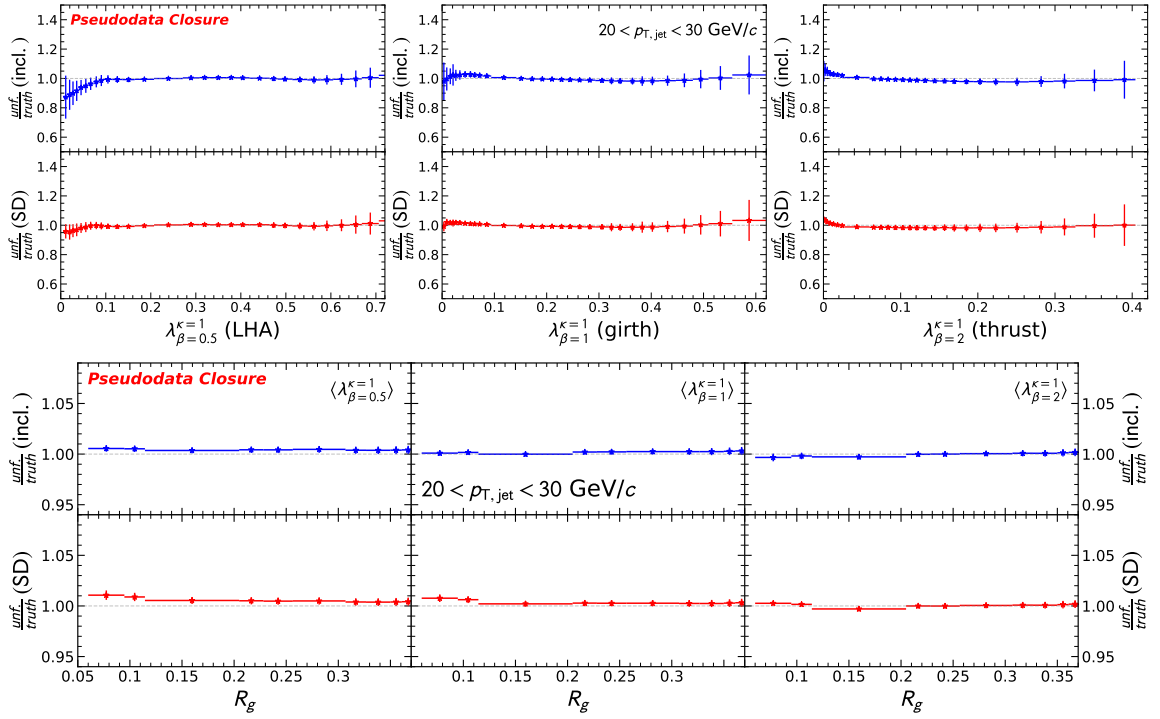


Figure 6.4: Closures for $\lambda_{\beta \in \{0.5, 1, 2\}}^{\kappa=1}$ distributions (top) and R_g profile (bottom) shown for jets with $20 \leq p_{T,\text{jet}} < 30 \text{ GeV}/c$.

6.1.3 Systematic Uncertainties

Jet Reconstruction Related Uncertainties

- Tracking Efficiency - The uncertainty in tracking efficiency is estimated at $\pm 4\%$ for $p + p$ events.

- Variation of Hadronic Correction - For the towers, we remove the full contribution of the track momentum for our central values. As a variation on that, we remove half of the total track momentum contribution.
- Tower Gain Uncertainty
- $p_{T,\text{jet}}$ resolution

Unfolding related uncertainties

- Variation of the regularization parameter(number of iterations) - For our central values, we have used 2 iterations as we found the result to converge reliably by that point. We vary the iterations between 3 and 5.

6.1.4 Results and Discussions

6.2 Angularities of Au+Au jets

6.2.1 Methodology

Charged-particle tracks and neutral energy depositions (towers) are measured using STAR's Time Projection Chamber (TPC) [34] and Barrel Electromagnetic Calorimeter (BEMC) [4] detectors respectively. Together, they provide full azimuthal coverage with a pseudo-rapidity acceptance of $|\eta| \leq 1$. The tracks and towers are clustered into jets using the anti- k_T algorithm with a jet resolution parameter $R = 0.4$, implemented using the FastJet library [35]. To suppress contributions of fake tracks and combinatorial background (especially in the context of the larger heavy-ion background), a “hard-core” constituent selection is applied, as was done in previous STAR analyses [36], which only allows tracks (towers) with $cp_{T,\text{trk}}(E_{T,\text{tow}}) \geq 2$ GeV to be clustered into jets. To enhance jet signal, only High-Tower (HT) triggered events, with at least one tower with $E_{T,\text{tower}} \geq 5.4$ GeV are considered. After clustering, only jets completely falling within acceptance ($|\eta_{\text{jet}}| \leq 0.6$) are kept. Jets with area, $A_{\text{jet}} < 0.4$ are rejected to further reduce the fake jet contribution.

Residual background and detector effects are removed by using a mapping between particle and detector level jets from an embedding simulation which involves PYTHIA-6 STAR tune [37] events processed into detector hits using GEANT3 [38] and added to real minimum-bias events from Au+Au collision environment. MultiFold [39] method was used to simultaneously unfold $p_{T,\text{jet}}$, η_{jet} , ϕ_{jet} , $N_{\text{cons,charged}}$, g , p_{T}^{D} and LeSub. MultiFold uses Dense Neural Networks (DNNs) implemented using the EnergyFlow package [40], which are trained on full embedding sample at the detector level and the generator level. Closure of the MultiFold is a measure of confidence in the algorithm determined by applying a pre-trained MultiFold on a detector level jet sample for which the particle level is known and comparing the two through ratios. Proper closure is demonstrated in Fig.6.5 for the observables studied in these proceedings, with the ratios of distributions being consistent with unity within statistical errors.

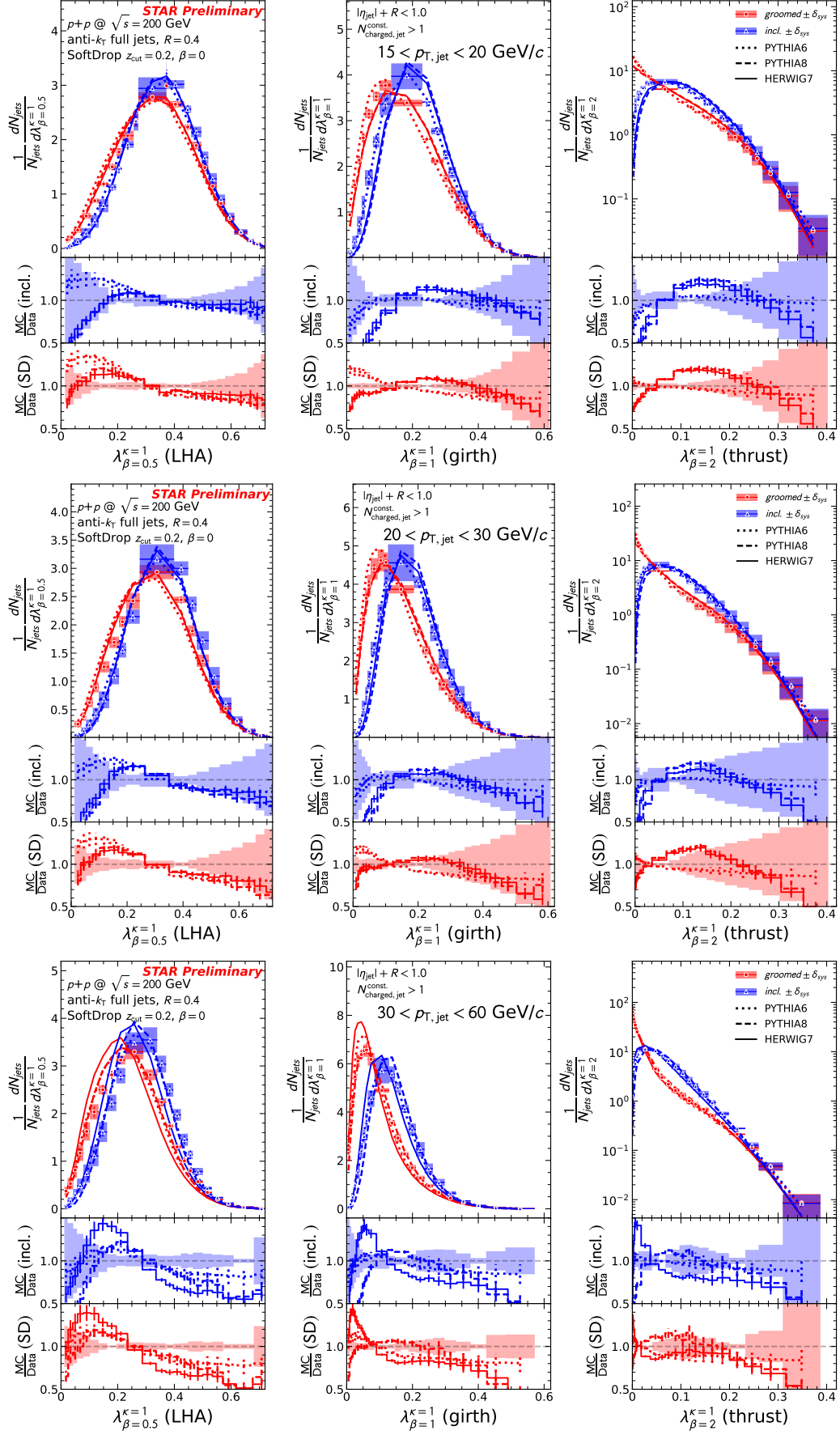
This study is the first instance of the MultiFold method being used to deconvolute detector effects from heavy-ion collision events.

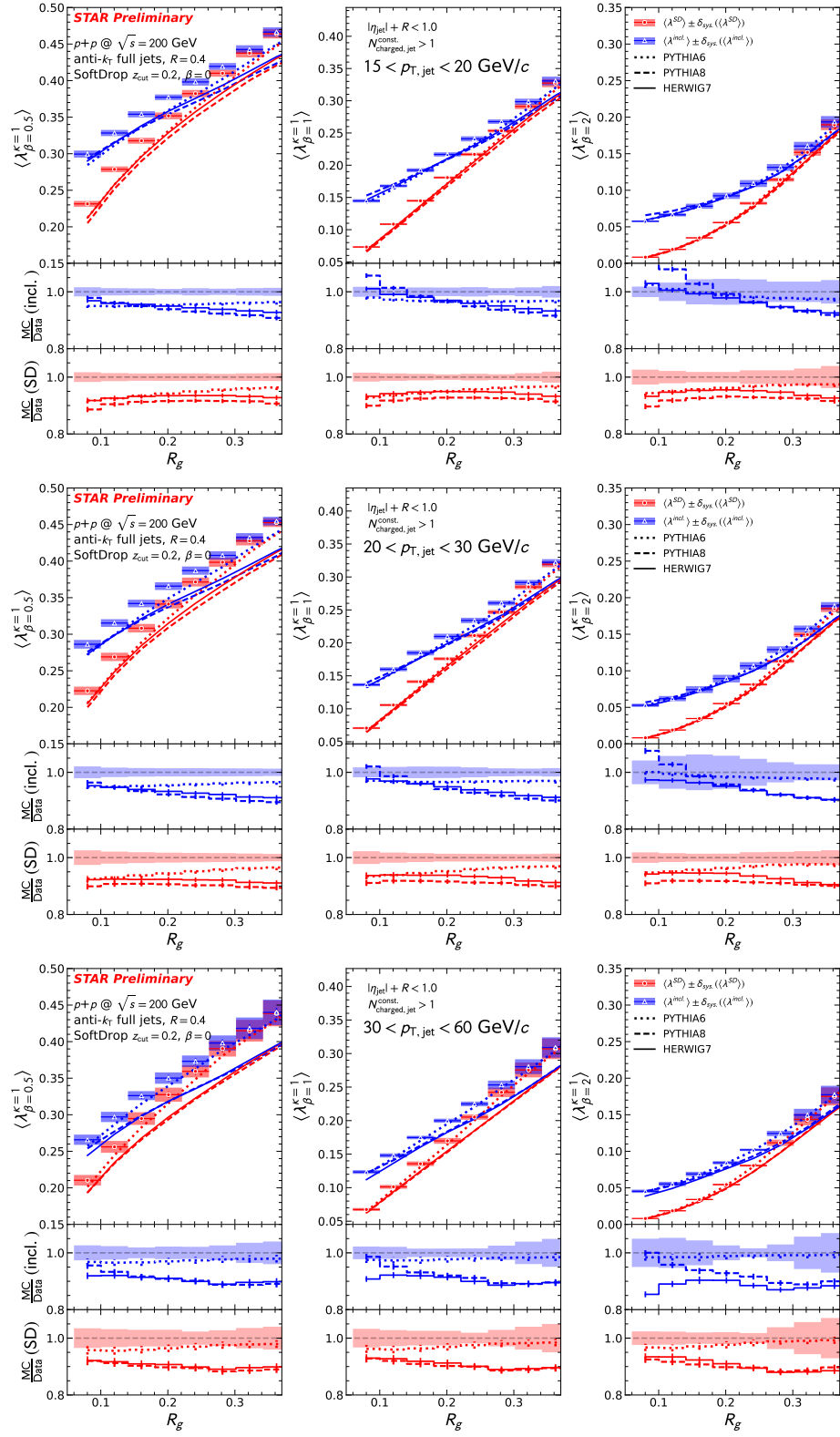
6.2.2 Result and Discussion

Fully corrected measurements of g , p_{T}^{D} and LeSub distributions calculated from jets with $p_{T,\text{jet}} > 20\text{GeV}/c$ from events within centrality ranges of 0-20% (central) and 40-80% (peripheral), and ratio comparisons between distributions from central and peripheral events are shown in Fig. 6.6. The p_{T}^{D} distribution shows a discontinuity at 0.7 due to the strong dependence on number of constituents in jet. Systematic uncertainties were assumed to be uncorrelated, with uncertainties from various sources being added in quadrature. Parameters varied for calculating the uncertainties in this study are the MultiFold regularization parameters of batch size and number of iterations. Any residual non-closure from MultiFold were also included in the calculation of systematic uncertainties. A prior variation uncertainty was added from previous calculations using STAR $\sqrt{s} = 200\text{ GeV}$ p+p collision data.

Results are consistent between central and peripheral collisions within conservative systematic uncertainties. This can be attributed to the assumption of uncorrelated systematic uncertainties, and a bias in the analysis towards hard-fragmented jets due to the selection of events and constituents for jet clustering. Harder-fragmented jets, on average are narrower and have lower interaction time with QGP, which could make any relative modification in central events compared to the peripheral ones lower than what would be potentially observed in an inclusive jet analysis.

6.3 Conclusion and Outlook





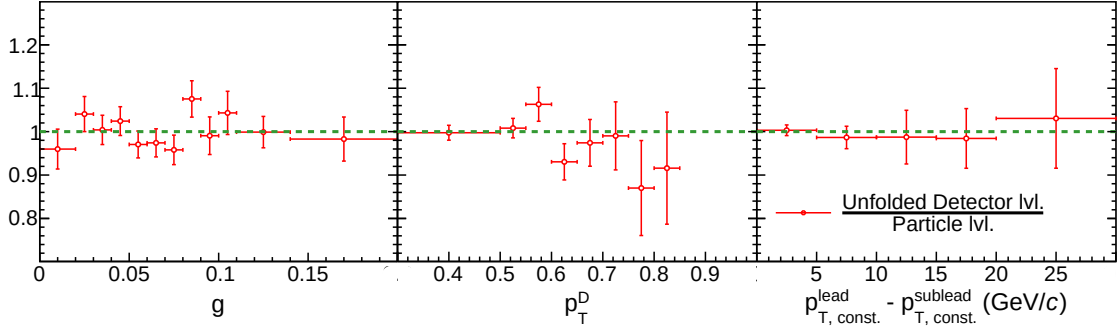


Figure 6.5: Closure test for g , p_T^D and LeSub showing the ratio of distributions calculated using jets from unfolded 0-20% centrality detector level and particle level events from Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV.

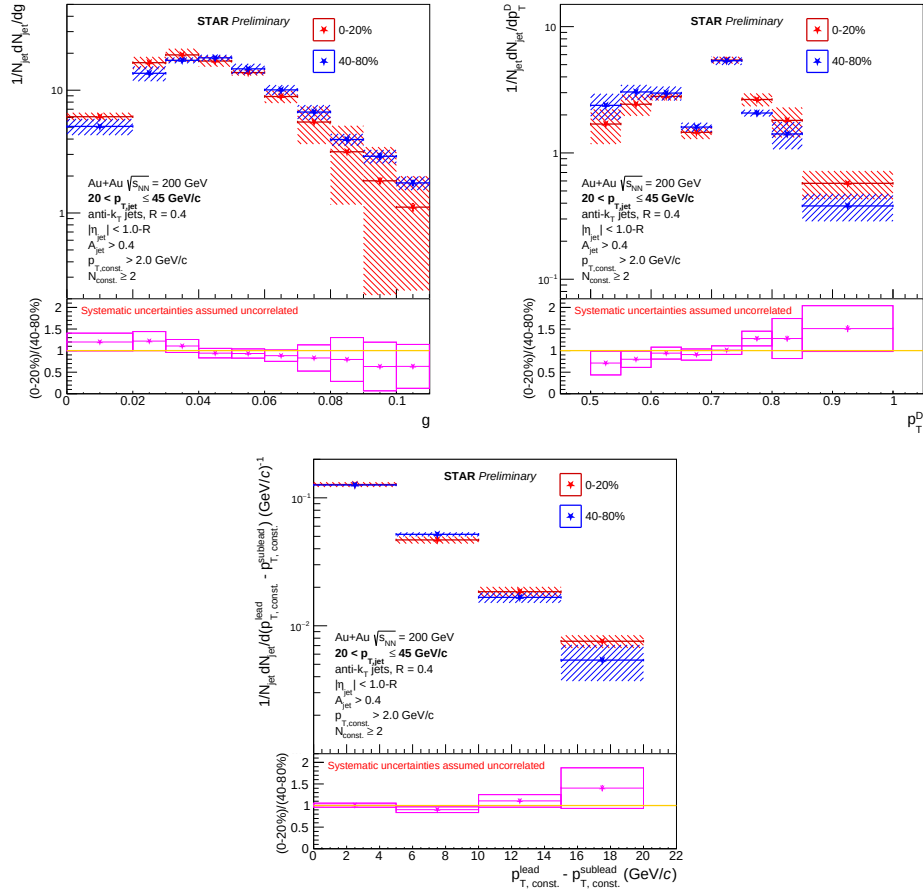


Figure 6.6: Normalized g , p_T^D and LeSub distributions for jets with $p_{T,jet} \geq 20$ GeV/c from central (0-20%, red stars) and peripheral (40-80%, blue stars) centrality ranges. Systematic uncertainties are shown as shaded boxes. Central over peripheral ratios (magenta stars) are in the lower panels with their systematic uncertainties represented as magenta boxes. Systematic uncertainties are assumed to be uncorrelated between the centrality ranges.

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Appendix A

Relativistic Kinematics

A.1 Laboratory vs Center-of-mass frames

A typical collision is a two-to-many reaction like:

$$\underbrace{A}_{\text{projectile}} + \underbrace{B}_{\text{target}} \rightarrow C_1 + C_2 + C_3 + \dots, \quad (\text{A.1})$$

In *laboratory frame*, the target is at rest, so,

$$\mathbf{p}_B^{\text{lab}} = \vec{0} \implies p_B^{\text{lab}} = (m_B, 0, 0, 0), \quad (\text{A.2})$$

In *center-of-mass* frame, the sum of three-momentum vectors vanish in the initial and final states

$$\mathbf{p}_A^{\text{cm}} + \mathbf{p}_B^{\text{cm}} = \sum_i \mathbf{p}_{C,i}^{\text{cm}} = \vec{0}, \quad (\text{A.3})$$

Using Lorentz invariance of the Mandelstam variable $s \equiv (p_A + p_B)_\mu (p_A + p_B)^\mu$:

$$\begin{aligned} s &= (E_A^{\text{lab}} + m_B)^2 - |\mathbf{p}_A^{\text{lab}}|^2 = (E_A^{\text{cm}} + E_B^{\text{cm}})^2 \\ \implies s &= m_A^2 + m_B^2 + 2m_B E_A^{\text{lab}} = (E^{\text{cm}})^2 \end{aligned} \quad (\text{A.4})$$

As relativistic energies ($\mathcal{O}(10^{2-3}\text{GeV})$) are much higher than reactant masses $\mathcal{O}(1 - 10\text{GeV})$,

$$\sqrt{s} = E^{\text{cm}} \approx \sqrt{2m_B E_A^{\text{lab}}} \quad (\text{A.5})$$

As the center-of-mass energy that produces new particles increases only as $\sqrt{E^{\text{lab}}}$, making any gain in particle production more expensive at relativistic energies. Thus modern colliders usually operate in center-of-mass frames.

A.2 Coordinate system

Considering the non-linear addition of velocities in relativistic mechanics,

$$\beta = \frac{\beta_1 + \beta_2}{1 + \beta_1 \beta_2}, \quad (\text{A.6})$$

Considering the transformation $y = \tanh^{-1} \beta$,

$$\begin{aligned} \tanh^{-1} \beta_1 + \tanh^{-1} \beta_2 &= \tanh^{-1} \frac{\beta_1 + \beta_2}{1 + \beta_1 \beta_2} = \tanh^{-1} \beta \\ \implies y_1 + y_2 &= y \end{aligned} \quad (\text{A.7})$$

The addition is linear again. We call this transformed velocity *rapidity*, given by:

$$y = \tanh^{-1} \beta = \frac{1}{2} \ln \frac{1 + \beta}{1 - \beta} = \frac{1}{2} \ln \frac{E + p_z}{E - p_z} \quad (\text{A.8})$$

Rapidity can grow without bound, and at small velocities, $y \approx \beta$, making it a relativistic analogue of velocity. Since it transforms linearly under Lorentz boosts, rapidity distribution of particles retain their shapes. For relativistic energies, $E \approx p$, making A.8,

$$y \approx \frac{1}{2} \ln \frac{p + p_z}{p - p_z} = -\ln \left(\tan \frac{\theta}{2} \right) = \eta \quad (\text{A.9})$$

η is called the *pseudo-rapidity*, the massless limit of rapidity. As shown in A.2, pseudo-rapidity can be directly determined from particle production angle θ measured with respect

to the beam axis.

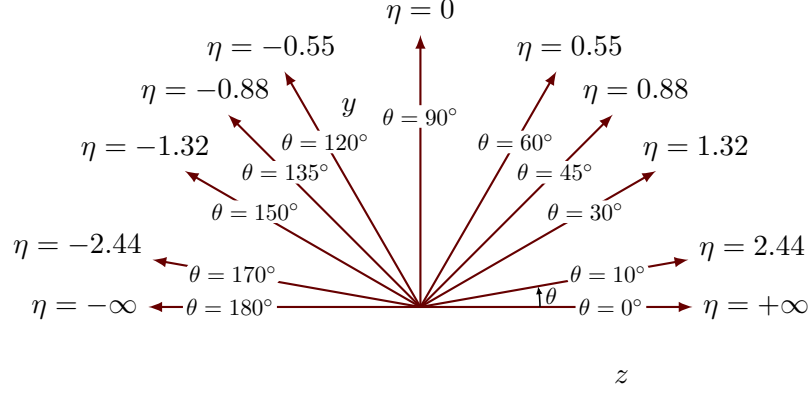


Figure A.1: η vs θ

Also considering *azimuthal angle* (φ) and *transverse-momentum* ($p_T = \sqrt{p_x^2 + p_y^2}$) of the particles, which are invariant under boosts along the beam-line by design, we can use (p_T, η, φ) as the coordinate system for the particles produced in the collision. Figure 2.6 illustrates this coordinate system for a cylindrical collider detector.